완벽하게 이해했습니다. 정리하면:

새 레포 `visualclimate-trojan` — 기존 Supabase(172,121행) 읽고 + 새 데이터 추가도 하고 — CLAUDE.md 완전 새로 — Claude가 분석/보고서/텍스트를 만들고, Tableau가 시각화하는 경계를 명확히 — 보고서 템플릿, Tableau 계산식 설계도까지 Claude가 생성할 수 있게.

지금부터 **Claude Code**에서 바로 실행할 수 있는 모든 파일을 만들어 드립니다. 파일별로 번호 매겨서, 하나도 빠짐없이 갑니다.

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visualclimate-trojan 전체 파일 세트

폴더 구조

```
visualclimate-trojan/
│
├── CLAUDE.md                          ← 파일 #1: 마스터 규칙서
│
├── data/
│   ├── source-registry.json           ← 파일 #2: 7개 소스 메타데이터
│   ├── citations.json                 ← 파일 #3: 핵심 인용 문헌
│   ├── schema.json                    ← 파일 #4: 23개 지표 스키마
│   ├── ndc-targets-v2.json            ← 파일 #5: NDC 3.0 포함 20국
│   ├── supabase-bridge.md             ← 파일 #6: 기존 Supabase 연결 가이드
│   └── tableau/
│       ├── countries_latest.csv       ← 파일 #7: Tableau 피드 (최신 지표)
│       ├── timeseries.csv             ← 파일 #8: Tableau 피드 (시계열)
│       ├── tableau-setup.md           ← 파일 #9: Tableau 연결/계산식 설계도
│       └── calculated-fields.md       ← 파일 #10: Tableau 계산 필드 전체 목록
│
├── .claude/
│   ├── skills/
│   │   ├── schema-explainer/SKILL.md      ← 파일 #11: 스킬 A1
│   │   ├── source-comparator/SKILL.md     ← 파일 #12: 스킬 A2
│   │   ├── data-quality-checker/SKILL.md  ← 파일 #13: 스킬 A3
│   │   ├── indicator-definer/SKILL.md     ← 파일 #14: 스킬 A4
│   │   ├── country-brief/SKILL.md         ← 파일 #15: 스킬 B1
│   │   ├── comparison-brief/SKILL.md      ← 파일 #16: 스킬 B2
│   │   ├── ndc-gap-brief/SKILL.md         ← 파일 #17: 스킬 B3
│   │   ├── qa-checker/SKILL.md            ← 파일 #18: 스킬 C1
│   │   ├── content-repurposer/SKILL.md    ← 파일 #19: 스킬 D1
│   │   └── tableau-spec-writer/SKILL.md   ← 파일 #20: 스킬 E1 (Tableau 설계)
│   └── agents/
```

```
│       └── engine-director.md          ← 파일 #21: 총괄 에이전트
│
├── templates/
│   ├── country-brief.md                ← 파일 #22: 보고서 템플릿
│   ├── comparison-brief.md             ← 파일 #23: 비교 템플릿
│   ├── ndc-gap-brief.md                ← 파일 #24: NDC 갭 템플릿
│   ├── linkedin-carousel.md            ← 파일 #25: 캐러셀 템플릿
│   └── tableau-dashboard-spec.md       ← 파일 #26: Tableau 대시보드 설계 템플릿
│
├── output/
│   └── .gitkeep                        ← 파일 #27
│
├── scripts/
│   ├── pull-from-supabase.ts           ← 파일 #28: 기존 DB에서 데이터 추출
│   └── push-to-supabase.ts             ← 파일 #29: 새 데이터 DB에 추가
│
├── docs/
│   ├── architecture.md                 ← 파일 #30: 시스템 아키텍처
│   ├── claude-vs-tableau.md            ← 파일 #31: 역할 분리 명세
│   └── content-strategy.md             ← 파일 #32: 정보성 vs 제품 구분
│
├── brand-config.md                     ← 파일 #33: 브랜드 설정
├── .env.example                        ← 파일 #34: 환경변수 템플릿
├── package.json                        ← 파일 #35: 의존성
└── README.md                           ← 파일 #36: 레포 설명
```

파일 #1: CLAUDE.md

```
# VisualClimate Trojan — CLAUDE.md v1.0 | 2026-04-07
# 이 파일이 프로젝트 내 모든 다른 문서보다 우선합니다.

## 이 프로젝트의 정체
VisualClimate(visualclimate.org)의 데이터 분석·보고서·콘텐츠 엔진.
기존 레포(Climate-RAG-for-ALL)는 웹앱 프론트엔드(Next.js + D3).
이 레포는 그 웹앱의 Supabase 데이터를 읽고, 새 데이터를 추가하고,
보고서를 자동 생성하고, Tableau 대시보드 설계도를 만들고,
LinkedIn 콘텐츠를 뽑아내는 백엔드 엔진.

## 아키텍처 3층 구조


Layer 3 — 출력물: 보고서 3종, Tableau 대시보드 3종, LinkedIn 콘텐츠
 Layer 2 — Claude 스킬 10개: 해석4 + 생성3 + 검증1 + 변환1 + Tableau1
 Layer 1 — 기반 자산: 이 파일, schema.json, source-registry.json, brand-config.md
```

데이터 소스 현황
- **기존 Supabase**: 172,121행 | 250국 | 61 지표 | 소스 6개
  - DB: countries(250), indicators(67), country_data(172,121)
  - 접속: .env의 SUPABASE_URL, SUPABASE_SERVICE_KEY
- **기존 레포 data/**: owid-energy-data.csv(9MB), ember-electricity.json(6국), climate-trace-ghg.json(6국), ndgain-scores.json, ndc-targets.json(20국, 2030만)
- **기존 레포 public/data/**: kaya/{ISO3}.json(68국), ndc-gap/{ISO3}.json(204국), risk-profile-{ISO3}.json(6국만)
- **이 레포에서 추가**: EDGAR, IRENA, NDC 3.0(2035), CAT ratings, Net Zero Tracker

기존 Supabase indicator_code 패턴 (반드시 참조)
- WB.* — World Bank WDI (경제, 인구, CO2)
- EMBER.* — Ember 전력 데이터
- OWID.* — Our World in Data 에너지
- NDGAIN.* — ND-GAIN 취약성/준비도
- CTRACE.* — Climate TRACE 부문별 배출 (POWER, TRANSPORTATION 등 9개)
- DERIVED.* — 파생 지표 (CO2_PER_GDP, ENERGY_TRANSITION 등)

Supabase에 새 지표 추가 시 이 패턴을 따를 것.
신규 패턴: EDGAR.*, IRENA.*, NDC.*, CAT.*, NETZERO.*

데이터 규칙 (모든 작업에 적용)

단위 표준
지표 유형	단위	비고
GHG 총배출	$MtCO_2eq$	국가 수준
1인당 배출	$tCO_2eq/capita$	
배출 원단위	$kgCO_2eq$/USD GDP PPP	2017 기준
발전량	TWh	
1차 에너지	EJ	
설비 용량	GW	
재생에너지 비중	% of total electricity generation	

GWP 기준
- 기본: IPCC AR5 (CH_4=28, N_2O=265)
- AR6 소스 → AR5 변환 + 각주 (CH_4: ×28/27.9)
- AR4 이하 → ⚠️ + 원본값과 AR5 환산값 병기
- 보고서 내 AR4/AR5/AR6 혼용 절대 금지

LULUCF
- 기본: 제외 (excluding LULUCF)
- 포함 시 "(incl. LULUCF)" 반드시 명시
- 같은 표에서 포함/제외 혼용 금지

출처 표기
- 모든 수치에 (소스명, 연도) 필수
- URL은 데이터 다운로드 직접 페이지
- 형식: (EDGAR 2025, Crippa et al.) 또는 (OWID, GCP 2025)

환각 방지
- data/ 또는 Supabase에 없는 수치 → 절대 생성 금지
- 모르면 [DATA MISSING] 표시
- 추정치 → "⚠️ 추정치" + 추정 방법 명시
- "약", "대략", "~" 최소화. 정확한 수치 우선.

비교 규칙
- 동일 소스, 동일 GWP, 동일 연도 강제
- 소스 간 수치 차이 > 10% → ⚠️ + 차이 원인 각주

보고서 출력 규칙
- 상단 필수: 국가명(영문+한국어), ISO3, Data vintage, Generated date
- templates/ 폴더의 템플릿 반드시 사용
- 출력 경로: output/{ISO3}-{report-type}-{YYYY-MM-DD}.md

Tableau 연동 규칙
- Claude가 만드는 CSV: data/tableau/countries_latest.csv, timeseries.csv
- 열 이름 고정: ISO3, country_name, year, indicator_code, value, unit, source
- Tableau 계산 필드 설계: data/tableau/calculated-fields.md에 문서화
- 새 대시보드 설계 시: templates/tableau-dashboard-spec.md 템플릿 사용

Claude가 하는 일 vs Tableau가 하는 일
Claude	Tableau
데이터 수집/정제 (ETL)	차트 렌더링
지표 계산/파생	대시보드 레이아웃
텍스트 분석/해석	인터랙티브 필터
보고서 마크다운 생성	지도 시각화
QA 검증	시계열/비교 차트
콘텐츠 변환	PDF 내보내기
Tableau 계산식 설계	계산식 실행
CSV 생성 (data/tableau/)	CSV 읽기 (데이터 소스)

프로젝트 파일 맵
작업	경로
기반 자산	CLAUDE.md, data/schema.json, data/source-registry.json
스킬	.claude/skills/{name}/SKILL.md (10개)
에이전트	.claude/agents/engine-director.md
보고서 템플릿	templates/ (5개)
생성된 출력물	output/
Tableau 피드	data/tableau/
Supabase 연동	scripts/pull-from-supabase.ts, push-to-supabase.ts
브랜드	brand-config.md
문서	docs/

금지사항

- 다크 테마/배경 금지 (기존 레포 디자인 시스템 유지)
- data/ 또는 Supabase에 없는 수치 생성 금지
- source-registry.json에 없는 소스 임의 인용 금지
- schema.json에 정의되지 않은 지표를 보고서에 넣지 않음
- 기존 Supabase 테이블 구조(countries, indicators, country_data) 변경 금지
- 기존 indicator_code 패턴 변경 금지 (신규 추가만 허용)

토큰 최적화
1. 이 파일의 "프로젝트 파일 맵"을 먼저 확인
2. schema.json → source-registry.json → 해당 국가 데이터 순으로 로드
3. 보고서 생성 시 템플릿 먼저 읽고, 데이터 채우기
4. QA는 보고서 생성 직후 반드시 실행

파일 #2: data/source-registry.json


```
{
  "registry_version": "1.0",
  "last_updated": "2026-04-07",
  "sources": {
    "EDGAR": {
      "full_name": "Emissions Database for Global Atmospheric Research",
      "publisher": "European Commission Joint Research Centre (JRC)",
      "url_data": "https://edgar.jrc.ec.europa.eu/dataset_ghg2025",
      "url_report": "https://edgar.jrc.ec.europa.eu/report_2025",
      "coverage_countries": 220,
      "coverage_years": "1970-2024",
      "gases": ["CO2", "CH4", "N2O", "F-gases"],
      "sectors": "IPCC 2006 categories (95 subsectors)",
      "methodology": "Tier 1 (activity data × default EF), top-down",
      "gwp": "AR5",
      "lulucf": "separate dataset (preliminary estimates)",
      "update_frequency": "annual (Sep-Oct)",
      "latest_data_year": 2024,
      "latest_release": "2025-09",
      "format": ["CSV", "Excel"],
      "license": "EU Open Data (public domain)",
      "strengths": "Most comprehensive country-sector-timeseries coverage globally",
      "limitations": "Tier 1 default emission factors; ±10-20% vs national inventories",
      "supabase_prefix": "EDGAR",
      "last_verified": "2026-04-07"
    },
    "CLIMATE_TRACE": {
      "full_name": "Climate TRACE",
      "publisher": "Climate TRACE Coalition",
      "url_data": "https://climatetrace.org/data",
      "url_api": "https://api.climatetrace.org/v7",
```

"coverage_countries": 250,
"coverage_years": "2015-2025",
"gases": ["CO2", "CH4", "N2O", "F-gases"],
"sectors": "67 subsectors (facility-level)",
"methodology": "Satellite + ML, bottom-up (facility level)",
"gwp": "AR5",
"lulucf": "included (forest fires, land-use change)",
"update_frequency": "monthly",
"latest_data_year": 2025,
"latest_release": "2026-02",
"format": ["CSV", "JSON API"],
"license": "CC-BY 4.0",
"strengths": "Facility-level data, monthly updates, most current",
"limitations": "No data before 2015, ML estimation uncertainty",
"supabase_prefix": "CTRACE",
"last_verified": "2026-04-07"
},
"OWID": {
"full_name": "Our World in Data — CO2 and GHG Emissions",
"publisher": "Global Change Data Lab (Oxford)",
"url_data": "https://github.com/owid/co2-data",
"url_csv": "https://owid-public.owid.io/data/co2/owid-co2-data.csv",
"coverage_countries": 200,
"coverage_years": "1750-2024",
"gases": ["CO2 (fossil+land use)", "CH4", "N2O", "GHG total"],
"underlying_sources": ["Global Carbon Project (CO2)", "EDGAR (CH4, N2O)", "CAIT/Climate Watch"],
"gwp": "varies by underlying source — CO2 N/A, CH4/N2O from EDGAR AR5",
"update_frequency": "annual (after GCP release)",
"latest_data_year": 2024,
"format": ["CSV", "JSON", "GitHub"],
"license": "CC-BY 4.0",
"strengths": "Longest timeseries (1750+), 79 variables, fully free",
"limitations": "Secondary processed data (not primary), no F-gases",
"supabase_prefix": "OWID",
"last_verified": "2026-04-07"
},
"EMBER": {
"full_name": "Ember Global Electricity Review",
"publisher": "Ember",
"url_data": "https://ember-climate.org/data/",
"coverage_countries": 215,
"coverage_years": "2000-2025",
"focus": "Electricity generation, capacity, carbon intensity of power",
"update_frequency": "monthly",
"latest_data_year": 2025,
"format": ["CSV", "API"],
"license": "CC-BY 4.0",

"strengths": "Fastest monthly power sector updates; detailed fuel breakdown",
"limitations": "Power sector only; no heat/transport/industry",
"supabase_prefix": "EMBER",
"last_verified": "2026-04-07"
},
"IRENA": {
"full_name": "International Renewable Energy Agency — Statistics",
"publisher": "IRENA",
"url_data": "https://www.irena.org/Data/Downloads",
"coverage_countries": 200,
"coverage_years": "2000-2024",
"focus": "Renewable energy capacity, generation, investment",
"update_frequency": "annual (March)",
"latest_data_year": 2024,
"format": ["CSV", "Excel", "API"],
"license": "Free (registration required)",
"strengths": "Authoritative RE source; solar/wind/hydro/bio breakdown",
"limitations": "RE only; no fossil fuel data",
"supabase_prefix": "IRENA",
"last_verified": "2026-04-07"
},
"NDGAIN": {
"full_name": "Notre Dame Global Adaptation Initiative Country Index",
"publisher": "University of Notre Dame",
"url_data": "https://gain.nd.edu/our-work/country-index/download-data/",
"coverage_countries": 181,
"coverage_years": "1995-2022",
"focus": "Climate vulnerability (6 sectors) + Adaptation readiness (3 components)",
"update_frequency": "annual (varies)",
"latest_data_year": 2022,
"format": ["CSV"],
"license": "Free",
"strengths": "Only comprehensive vulnerability+readiness index at country level",
"limitations": "2-3 year lag; 181 countries (missing small islands)",
"supabase_prefix": "NDGAIN",
"last_verified": "2026-04-07"
},
"CLIMATE_WATCH": {
"full_name": "Climate Watch — World Resources Institute",
"publisher": "WRI",
"url_data": "https://www.climatewatchdata.org/data-explorer",
"url_ndc": "https://www.climatewatchdata.org/2020-ndc-tracker",
"url_api": "https://www.climatewatchdata.org/api/v1/",
"coverage_countries": 194,
"focus": "NDC targets, historical emissions (CAIT/PIK PRIMAP), climate policy",
"update_frequency": "continuous (NDC submissions)",
"format": ["CSV", "JSON API"],
"license": "CC-BY 4.0",

"strengths": "Best structured NDC target data; policy linkage; SDG mapping",
"limitations": "Emissions data is secondary (from CAIT/PIK)",
"supabase_prefix": "NDC",
"last_verified": "2026-04-07"
},
"WORLD_BANK_WDI": {
"full_name": "World Development Indicators",
"publisher": "World Bank",
"url_data": "https://databank.worldbank.org/source/world-development-indicators",
"url_api": "https://api.worldbank.org/v2/",
"coverage_countries": 266,
"coverage_years": "1960-2024",
"focus": "GDP, population, income groups, regional classification, basic climate indicators",
"update_frequency": "quarterly",
"format": ["CSV", "JSON API"],
"license": "CC-BY 4.0",
"key_indicators": {
"EN.ATM.CO2E.PC": "CO2 per capita (metric tons)",
"EN.ATM.CO2E.KT": "CO2 total (kt)",
"EN.ATM.GHGT.KT.CE": "Total GHG (kt CO2eq)",
"EG.FEC.RNEW.ZS": "Renewable energy share (%)",
"SP.POP.TOTL": "Population",
"NY.GDP.MKTP.PP.CD": "GDP PPP (current intl $)",
"NY.GDP.PCAP.CD": "GDP per capita (current USD)"
},
"strengths": "Most countries (266); longest economic timeseries; official classifications",
"limitations": "Climate data 2-3yr lag; CO2 only (not full GHG from WB directly)",
"supabase_prefix": "WB",
"last_verified": "2026-04-07"
}
}
}

파일 #3: data/citations.json

{
"citations_version": "1.0",
"last_updated": "2026-04-07",
"citations": {
"IPCC_AR6_WG1": {
"title": "Climate Change 2021: The Physical Science Basis",
"authors": "IPCC Working Group I",
"year": 2021,
"publisher": "Cambridge University Press",
"doi": "10.1017/9781009157896",

```
  "url": "https://www.ipcc.ch/report/ar6/wg1/",
  "use_in_project": "GWP values (Table 7.15), warming projections, carbon budget",
  "key_facts": {
    "gwp_ch4_100yr": 27.9,
    "gwp_n2o_100yr": 273,
    "remaining_carbon_budget_1_5C_500Gt": "500 GtCO2 from 2020 (50% probability)"
  }
},
"IPCC_AR6_WG3": {
  "title": "Climate Change 2022: Mitigation of Climate Change",
  "authors": "IPCC Working Group III",
  "year": 2022,
  "publisher": "Cambridge University Press",
  "doi": "10.1017/9781009157926",
  "url": "https://www.ipcc.ch/report/ar6/wg3/",
  "use_in_project": "Sectoral mitigation pathways, 1.5°C/2°C scenarios, technology costs",
  "key_facts": {
    "global_ghg_2019": "59 GtCO2eq (AR6 GWP)",
    "1_5C_requires": "43% reduction by 2030 vs 2019",
    "2C_requires": "27% reduction by 2030 vs 2019"
  }
},
"IPCC_AR5_GWP": {
  "title": "Climate Change 2013: The Physical Science Basis — Table 8.7",
  "authors": "IPCC Working Group I",
  "year": 2013,
  "url": "https://www.ipcc.ch/report/ar5/wg1/",
  "use_in_project": "Standard GWP basis for this project (AR5)",
  "key_facts": {
    "gwp_ch4_100yr": 28,
    "gwp_n2o_100yr": 265,
    "gwp_sf6_100yr": 23500
  }
},
"UNEP_EGR_2024": {
  "title": "Emissions Gap Report 2024: No More Hot Air — Please!",
  "authors": "UNEP",
  "year": 2024,
  "url": "https://www.unep.org/resources/emissions-gap-report-2024",
  "use_in_project": "Global emissions gap, NDC ambition assessment, G20 analysis",
  "key_facts": {
    "global_ghg_2023": "57.1 GtCO2eq (excl. LULUCF)",
    "ndc_gap_2030": "Unconditional NDCs leave 20-23 GtCO2eq gap for 1.5°C",
    "g20_share": "G20 accounts for ~77% of global GHG"
  }
},
"WMO_STATE_2024": {
  "title": "WMO State of the Global Climate 2024",
```

"authors": "World Meteorological Organization",
"year": 2025,
"url": "https://wmo.int/publication/state-of-global-climate-2024",
"use_in_project": "Temperature records, climate indicators, extreme events",
"key_facts": {
"global_mean_temp_2024": "1.55°C above pre-industrial (1850-1900)",
"warmest_year": "2024 confirmed warmest on record",
"ocean_heat_content": "Record high in 2024"
}
},
"EDGAR_2025": {
"title": "EDGAR — Emissions Database for Global Atmospheric Research, GHG emissions of all world countries, 2025 Report",
"authors": "Crippa, M., Guizzardi, D., et al.",
"year": 2025,
"publisher": "European Commission, JRC",
"doi": "10.2760/812002",
"url": "https://edgar.jrc.ec.europa.eu/report_2025",
"use_in_project": "Primary source for historical GHG emissions by country/sector",
"key_facts": {
"global_ghg_2024": "53.2 GtCO2eq (excl. LULUCF, AR5 GWP)",
"co2_fossil_2024": "37.4 GtCO2"
}
},
"GCP_2024": {
"title": "Global Carbon Budget 2024",
"authors": "Friedlingstein, P., et al.",
"year": 2024,
"doi": "10.5194/essd-16-5567-2024",
"url": "https://globalcarbonbudget.org/",
"use_in_project": "CO2 fossil emissions, land-use emissions, carbon sinks",
"key_facts": {
"co2_fossil_2024": "37.4 GtCO2",
"co2_lulucf_2024": "4.2 GtCO2",
"atmospheric_growth_2024": "2.8 ppm"
}
},
"CAT_METHODOLOGY": {
"title": "Climate Action Tracker — Methodology",
"authors": "Climate Analytics & NewClimate Institute",
"year": 2024,
"url": "https://climateactiontracker.org/methodology/",
"use_in_project": "Country NDC ratings, fair share assessment",
"key_facts": {
"rating_scale": ["1.5C compatible", "Almost sufficient", "Insufficient", "Highly insufficient", "Critically insufficient"],
"countries_covered": 40
}

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    },
    "NDGAIN_METHODOLOGY": {
      "title": "ND-GAIN Country Index Technical Report",
      "authors": "University of Notre Dame",
      "year": 2023,
      "url": "https://gain.nd.edu/our-work/country-index/methodology/",
      "use_in_project": "Vulnerability/readiness scoring methodology",
      "key_facts": {
        "vulnerability_sectors": ["food", "water", "health", "ecosystem", "habitat",
"infrastructure"],
        "readiness_components": ["economic", "governance", "social"],
        "score_range": "0-100 (higher=better prepared)"
      }
    },
    "PARIS_AGREEMENT": {
      "title": "Paris Agreement",
      "authors": "UNFCCC",
      "year": 2015,
      "url": "https://unfccc.int/process-and-meetings/the-paris-agreement",
      "use_in_project": "1.5°C/2°C targets, NDC framework, Article references",
      "key_facts": {
        "temperature_goal": "Well below 2°C, pursuing 1.5°C",
        "ndc_cycle": "Every 5 years with progression",
        "ndc3_deadline": "February 2025 (10 years from NDC2)"
      }
    }
  }
}
```

파일 #4: data/schema.json


```
{
  "schema_version": "1.0",
  "last_updated": "2026-04-07",
  "description": "23 core indicators for the Climate Intelligence Engine. Maps to existing
Supabase indicator_codes where available.",
  "indicator_groups": {
    "emissions": {
      "total_ghg_excl_lulucf": {
        "label": "Total GHG emissions (excl. LULUCF)",
        "unit": "MtCO2eq",
        "gwp": "AR5",
        "primary_source": "EDGAR",
        "secondary_source": "CLIMATE_TRACE",
        "supabase_code": "WB.EN.ATM.GHGT.KT.CE",
        "supabase_unit_conversion": "kt_to_Mt: value / 1000",
```

"cross_check_threshold_pct": 10,
"tableau_field": "total_ghg",
"notes": "WB stores in kt CO2eq. Convert to Mt for reports. EDGAR and CT cross-check."
},
"co2_fossil": {
"label": "CO2 from fossil fuels and industry",
"unit": "MtCO2",
"primary_source": "EDGAR",
"secondary_source": "OWID",
"supabase_code": "WB.EN.ATM.CO2E.KT",
"supabase_unit_conversion": "kt_to_Mt: value / 1000",
"tableau_field": "co2_fossil"
},
"ch4_total": {
"label": "Total methane emissions",
"unit": "MtCO2eq",
"gwp": "AR5",
"primary_source": "EDGAR",
"supabase_code": null,
"tableau_field": "ch4_total",
"notes": "Not in current Supabase. Add via EDGAR ETL. CH4 in CO2eq = CH4(Mt) × 28"
},
"ghg_per_capita": {
"label": "GHG per capita",
"unit": "tCO2eq/capita",
"derived_from": ["total_ghg_excl_lulucf", "population"],
"formula": "total_ghg_excl_lulucf * 1000000 / population",
"supabase_code": "WB.EN.ATM.CO2E.PC",
"supabase_note": "WB has CO2/capita only, not full GHG/capita. Derive from total_ghg + pop.",
"tableau_field": "ghg_per_capita",
"tableau_calc": "[total_ghg] * 1000000 / [population]"
},
"ghg_intensity": {
"label": "GHG intensity of GDP",
"unit": "kgCO2eq/USD GDP PPP",
"derived_from": ["total_ghg_excl_lulucf", "gdp_ppp"],
"formula": "total_ghg_excl_lulucf * 1e9 / gdp_ppp",
"supabase_code": "DERIVED.CO2_PER_GDP",
"tableau_field": "ghg_intensity",
"tableau_calc": "[total_ghg] * 1000000000 / [gdp_ppp]"
},
"sector_energy_pct": {
"label": "Energy sector share of total GHG",
"unit": "%",
"primary_source": "EDGAR",

"supabase_code": "CTRACE.POWER",
"supabase_note": "CTRACE has power sector only. EDGAR has full energy sector.",
"tableau_field": "sector_energy_pct"
},
"sector_industry_pct": {
"label": "Industrial processes share",
"unit": "%",
"primary_source": "EDGAR",
"supabase_code": "CTRACE.MANUFACTURING",
"tableau_field": "sector_industry_pct"
},
"sector_agriculture_pct": {
"label": "Agriculture share",
"unit": "%",
"primary_source": "EDGAR",
"supabase_code": "CTRACE.AGRICULTURE",
"tableau_field": "sector_agriculture_pct"
},
"sector_waste_pct": {
"label": "Waste share",
"unit": "%",
"primary_source": "EDGAR",
"supabase_code": "CTRACE.WASTE",
"tableau_field": "sector_waste_pct"
},
"ghg_10yr_trend_pct": {
"label": "10-year GHG emission trend",
"unit": "%",
"derived_from": ["total_ghg_excl_lulucf"],
"formula": "((value_latest / value_10yr_prior) - 1) * 100",
"supabase_code": null,
"tableau_field": "ghg_10yr_trend",
"tableau_calc": "(LOOKUP([total_ghg], 0) / LOOKUP([total_ghg], -10) - 1) * 100"
}
},
"energy": {
"renewable_share_electricity": {
"label": "Renewable share of electricity generation",
"unit": "%",
"primary_source": "IRENA",
"secondary_source": "EMBER",
"supabase_code": "WB.EG.FEC.RNEW.ZS",
"supabase_note": "WB has renewable share of final energy, not electricity. IRENA/Ember preferred.",
"tableau_field": "re_share_elec"
},
"fossil_share_primary_energy": {
"label": "Fossil fuel share of primary energy",

```
    "unit": "%",
    "primary_source": "OWID",
    "supabase_code": null,
    "tableau_field": "fossil_share_pe",
    "notes": "Available in data/owid-energy-data.csv column: fossil_share_energy"
  },
  "coal_phaseout_status": {
    "label": "Coal phase-out commitment status",
    "unit": "text",
    "primary_source": "Global Coal Plant Tracker",
    "values": ["committed", "under_discussion", "no_commitment", "expanding"],
    "supabase_code": null,
    "tableau_field": "coal_phaseout"
  }
},
"policy": {
  "ndc30_submitted": {
    "label": "NDC 3.0 (2025 round) submitted",
    "unit": "boolean",
    "primary_source": "UNFCCC NDC Registry",
    "supabase_code": null,
    "tableau_field": "ndc30_submitted"
  },
  "ndc_2030_target": {
    "label": "2030 NDC target description",
    "unit": "text",
    "primary_source": "CLIMATE_WATCH",
    "supabase_code": null,
    "tableau_field": null,
    "notes": "Text field — not for Tableau chart, used in reports only"
  },
  "ndc_2035_target": {
    "label": "2035 NDC target description",
    "unit": "text",
    "primary_source": "UNFCCC NDC Registry",
    "supabase_code": null,
    "tableau_field": null,
    "notes": "NDC 3.0 specific. Text field for reports."
  },
  "net_zero_target_year": {
    "label": "Net-zero target year",
    "unit": "year",
    "primary_source": "Net Zero Tracker",
    "supabase_code": null,
    "tableau_field": "nz_year"
  },
  "net_zero_legal_status": {
    "label": "Legal status of net-zero target",
```

```
      "unit": "text",
      "values": ["in_law", "in_policy", "pledge", "none"],
      "primary_source": "Net Zero Tracker",
      "supabase_code": null,
      "tableau_field": "nz_legal"
    },
    "cat_rating": {
      "label": "Climate Action Tracker overall rating",
      "unit": "text",
      "primary_source": "CAT",
      "values": ["1.5C_compatible", "Almost_sufficient", "Insufficient", "Highly_insufficient",
"Critically_insufficient"],
      "supabase_code": null,
      "tableau_field": "cat_rating",
      "notes": "CAT covers ~40 countries only. Others = [NOT RATED]"
    }
  },
  "vulnerability": {
    "ndgain_rank": {
      "label": "ND-GAIN Country Index rank",
      "unit": "rank (1=best)",
      "primary_source": "NDGAIN",
      "supabase_code": "NDGAIN.RANK",
      "tableau_field": "ndgain_rank"
    },
    "ndgain_vulnerability": {
      "label": "ND-GAIN Vulnerability score",
      "unit": "score (0-1, higher=more vulnerable)",
      "primary_source": "NDGAIN",
      "supabase_code": "NDGAIN.VULNERABILITY",
      "tableau_field": "ndgain_vuln"
    },
    "ndgain_readiness": {
      "label": "ND-GAIN Readiness score",
      "unit": "score (0-1, higher=more ready)",
      "primary_source": "NDGAIN",
      "supabase_code": "NDGAIN.READINESS",
      "tableau_field": "ndgain_ready"
    }
  },
  "context": {
    "population": {
      "label": "Population",
      "unit": "persons",
      "primary_source": "WORLD_BANK_WDI",
      "supabase_code": "WB.SP.POP.TOTL",
      "tableau_field": "population"
    },
```

```
      "gdp_ppp": {
        "label": "GDP (PPP, current international $)",
        "unit": "USD",
        "primary_source": "WORLD_BANK_WDI",
        "supabase_code": "WB.NY.GDP.MKTP.PP.CD",
        "tableau_field": "gdp_ppp"
      },
      "income_group": {
        "label": "World Bank income group",
        "unit": "text",
        "values": ["High income", "Upper middle income", "Lower middle income", "Low
income"],
        "primary_source": "WORLD_BANK_WDI",
        "supabase_table": "countries",
        "supabase_column": "income_group",
        "tableau_field": "income_group"
      },
      "region": {
        "label": "World Bank region",
        "unit": "text",
        "primary_source": "WORLD_BANK_WDI",
        "supabase_table": "countries",
        "supabase_column": "region",
        "tableau_field": "region"
      }
    }
  }
}
```

파일 #5: data/ndc-targets-v2.json


```
{
  "metadata": {
    "version": "2.0",
    "generated_at": "2026-04-07",
    "source": "UNFCCC NDC Registry + Climate Watch + Climate Action Tracker",
    "total_countries": 20,
    "changes_from_v1": "Added NDC 3.0 fields (target_year_2035,
target_pct_reduction_2035, ndc3_submitted, ndc3_submission_date)"
  },
  "targets": {
    "KOR": {
      "country_name": "South Korea",
      "ndc_2030": {
        "target_year": 2030,
        "target_pct_reduction": 40,
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"reference_year": 2018,
"reference_type": "absolute",
"reference_emissions_MtCO2eq": 727.6,
"target_emissions_MtCO2eq": 436.6,
"conditional": false,
"submission_status": "updated",
"notes": "Updated NDC (2021): 40% reduction from 2018 levels by 2030"
},
"ndc_2035": {
"ndc3_submitted": false,
"target_year": 2035,
"target_pct_reduction": null,
"reference_year": null,
"reference_type": null,
"ndc3_submission_date": null,
"notes": "NDC 3.0 not yet submitted as of 2026-04-07. Expected to align with net-zero 2050."
},
"net_zero": {
"target_year": 2050,
"legal_status": "in_law",
"law_name": "Carbon Neutrality and Green Growth Framework Act (2021)"
},
"cat_rating": "Highly_insufficient"
},
"USA": {
"country_name": "United States",
"ndc_2030": {
"target_year": 2030,
"target_pct_reduction": 50,
"reference_year": 2005,
"reference_type": "absolute",
"reference_emissions_MtCO2eq": 7417,
"target_emissions_MtCO2eq": 3709,
"conditional": false,
"submission_status": "updated",
"notes": "NDC (2021): 50-52% reduction from 2005 levels by 2030; using lower bound 50%"
},
"ndc_2035": {
"ndc3_submitted": false,
"target_year": 2035,
"target_pct_reduction": null,
"reference_year": null,
"reference_type": null,
"ndc3_submission_date": null,
"notes": "NDC 3.0 status uncertain under current administration."
},

"net_zero": {
"target_year": 2050,
"legal_status": "pledge",
"law_name": null
},
"cat_rating": "Insufficient"
},
"DEU": {
"country_name": "Germany",
"ndc_2030": {
"target_year": 2030,
"target_pct_reduction": 65,
"reference_year": 1990,
"reference_type": "absolute",
"reference_emissions_MtCO2eq": 1251,
"target_emissions_MtCO2eq": 438,
"conditional": false,
"submission_status": "updated",
"notes": "German Climate Protection Act (2021): 65% from 1990; part of EU NDC (55% EU-wide)"
},
"ndc_2035": {
"ndc3_submitted": true,
"target_year": 2035,
"target_pct_reduction": 78,
"reference_year": 1990,
"reference_type": "absolute",
"ndc3_submission_date": "2025-02-14",
"notes": "EU NDC 3.0 (Feb 2025): 90% net reduction by 2040 (EU-wide). Germany national: 78% by 2035 (Climate Protection Act)."
},
"net_zero": {
"target_year": 2045,
"legal_status": "in_law",
"law_name": "Federal Climate Change Act (Bundes-Klimaschutzgesetz), amended 2021"
},
"cat_rating": "Insufficient"
},
"CHN": {
"country_name": "China",
"ndc_2030": {
"target_year": 2030,
"target_pct_reduction": 65,
"reference_year": 2005,
"reference_type": "intensity",
"conditional": false,
"submission_status": "updated",

"notes": "Updated NDC (2021): 65% carbon intensity reduction from 2005 by 2030; peak before 2030"
},
"ndc_2035": {
"ndc3_submitted": false,
"target_year": 2035,
"target_pct_reduction": null,
"reference_year": null,
"reference_type": null,
"ndc3_submission_date": null,
"notes": "NDC 3.0 not yet submitted. Expected intensity-based target."
},
"net_zero": {
"target_year": 2060,
"legal_status": "pledge",
"law_name": null
},
"cat_rating": "Highly_insufficient"
},
"IND": {
"country_name": "India",
"ndc_2030": {
"target_year": 2030,
"target_pct_reduction": 45,
"reference_year": 2005,
"reference_type": "intensity",
"conditional": false,
"submission_status": "updated",
"notes": "Updated NDC (2022): 45% carbon intensity reduction from 2005 by 2030"
},
"ndc_2035": {
"ndc3_submitted": false,
"target_year": 2035,
"target_pct_reduction": null,
"reference_year": null,
"reference_type": null,
"ndc3_submission_date": null,
"notes": "NDC 3.0 not yet submitted."
},
"net_zero": {
"target_year": 2070,
"legal_status": "pledge",
"law_name": null
},
"cat_rating": "Highly_insufficient"
},
"JPN": {
"country_name": "Japan",

"ndc_2030": {
"target_year": 2030,
"target_pct_reduction": 46,
"reference_year": 2013,
"reference_type": "absolute",
"conditional": false,
"submission_status": "updated",
"notes": "Updated NDC (2021): 46% reduction from 2013 levels by 2030"
},
"ndc_2035": {
"ndc3_submitted": true,
"target_year": 2035,
"target_pct_reduction": 60,
"reference_year": 2013,
"reference_type": "absolute",
"ndc3_submission_date": "2025-02-04",
"notes": "NDC 3.0 (Feb 2025): 60% reduction from 2013 levels by 2035"
},
"net_zero": {
"target_year": 2050,
"legal_status": "in_law",
"law_name": "Act on Promotion of Global Warming Countermeasures (amended 2021)"
},
"cat_rating": "Insufficient"
},
"GBR": {
"country_name": "United Kingdom",
"ndc_2030": {
"target_year": 2030,
"target_pct_reduction": 68,
"reference_year": 1990,
"reference_type": "absolute",
"conditional": false,
"submission_status": "updated",
"notes": "NDC (2020): 68% reduction from 1990 levels by 2030"
},
"ndc_2035": {
"ndc3_submitted": true,
"target_year": 2035,
"target_pct_reduction": 81,
"reference_year": 1990,
"reference_type": "absolute",
"ndc3_submission_date": "2025-02-04",
"notes": "NDC 3.0 (Feb 2025): 81% reduction from 1990 by 2035 (aligned with 6th Carbon Budget)"
},
"net_zero": {
"target_year": 2050,

"legal_status": "in_law",
"law_name": "Climate Change Act 2008 (amended 2019)"
},
"cat_rating": "Insufficient"
},
"BRA": {
"country_name": "Brazil",
"ndc_2030": {
"target_year": 2030,
"target_pct_reduction": 50,
"reference_year": 2005,
"reference_type": "absolute",
"conditional": false,
"submission_status": "updated",
"notes": "Updated NDC (2023): 50% reduction from 2005 levels by 2030"
},
"ndc_2035": {
"ndc3_submitted": true,
"target_year": 2035,
"target_pct_reduction": 67,
"reference_year": 2005,
"reference_type": "absolute",
"ndc3_submission_date": "2025-02-10",
"notes": "NDC 3.0 (Feb 2025): 59-67% reduction from 2005 by 2035; using upper bound 67%"
},
"net_zero": {
"target_year": 2050,
"legal_status": "in_policy",
"law_name": null
},
"cat_rating": "Insufficient"
},
"CAN": {
"country_name": "Canada",
"ndc_2030": {
"target_year": 2030,
"target_pct_reduction": 40,
"reference_year": 2005,
"reference_type": "absolute",
"conditional": false,
"submission_status": "updated",
"notes": "Updated NDC (2021): 40-45% from 2005 by 2030; using lower bound"
},
"ndc_2035": {
"ndc3_submitted": false,
"target_year": 2035,
"target_pct_reduction": null,

```
    "reference_year": null,
    "reference_type": null,
    "ndc3_submission_date": null,
    "notes": "NDC 3.0 not yet submitted."
  },
  "net_zero": {
    "target_year": 2050,
    "legal_status": "in_law",
    "law_name": "Canadian Net-Zero Emissions Accountability Act (2021)"
  },
  "cat_rating": "Insufficient"
},
"AUS": {
  "country_name": "Australia",
  "ndc_2030": {
    "target_year": 2030,
    "target_pct_reduction": 43,
    "reference_year": 2005,
    "reference_type": "absolute",
    "conditional": false,
    "submission_status": "updated",
    "notes": "Updated NDC (2022): 43% reduction from 2005 by 2030"
  },
  "ndc_2035": {
    "ndc3_submitted": false,
    "target_year": 2035,
    "target_pct_reduction": null,
    "reference_year": null,
    "reference_type": null,
    "ndc3_submission_date": null,
    "notes": "NDC 3.0 not yet submitted."
  },
  "net_zero": {
    "target_year": 2050,
    "legal_status": "in_law",
    "law_name": "Climate Change Act 2022"
  },
  "cat_rating": "Insufficient"
},
"FRA": {
  "country_name": "France",
  "ndc_2030": {
    "target_year": 2030,
    "target_pct_reduction": 55,
    "reference_year": 1990,
    "reference_type": "absolute",
    "conditional": false,
    "submission_status": "updated",
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"notes": "EU NDC (2020): 55% net reduction from 1990 (EU-wide)"
},
"ndc_2035": {
"ndc3_submitted": true,
"target_year": 2035,
"target_pct_reduction": null,
"reference_year": null,
"reference_type": null,
"ndc3_submission_date": "2025-02-14",
"notes": "Part of EU NDC 3.0: 90% net reduction by 2040 (EU-wide). National 2035 breakdown TBD."
},
"net_zero": {
"target_year": 2050,
"legal_status": "in_law",
"law_name": "Loi Énergie-Climat (2019)"
},
"cat_rating": "Insufficient"
},
"IDN": {
"country_name": "Indonesia",
"ndc_2030": {
"target_year": 2030,
"target_pct_reduction": 32,
"reference_year": null,
"reference_type": "bau",
"conditional": false,
"submission_status": "updated",
"notes": "Enhanced NDC (2022): 31.89% unconditional below BAU by 2030"
},
"ndc_2035": {
"ndc3_submitted": false,
"target_year": 2035,
"target_pct_reduction": null,
"reference_year": null,
"reference_type": null,
"ndc3_submission_date": null,
"notes": "NDC 3.0 not yet submitted."
},
"net_zero": {
"target_year": 2060,
"legal_status": "in_policy",
"law_name": null
},
"cat_rating": "Highly_insufficient"
},
"NGA": {
"country_name": "Nigeria",

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    "ndc_2030": {
      "target_year": 2030,
      "target_pct_reduction": 47,
      "reference_year": null,
      "reference_type": "bau",
      "conditional": false,
      "submission_status": "submitted",
      "notes": "NDC (2021): 47% unconditional below BAU by 2030"
    },
    "ndc_2035": {
      "ndc3_submitted": false,
      "target_year": 2035,
      "target_pct_reduction": null,
      "reference_year": null,
      "reference_type": null,
      "ndc3_submission_date": null,
      "notes": "NDC 3.0 not yet submitted."
    },
    "net_zero": {
      "target_year": 2060,
      "legal_status": "pledge",
      "law_name": null
    },
    "cat_rating": "Highly_insufficient"
  },
  "BGD": {
    "country_name": "Bangladesh",
    "ndc_2030": {
      "target_year": 2030,
      "target_pct_reduction": 22,
      "reference_year": null,
      "reference_type": "bau",
      "conditional": false,
      "submission_status": "updated",
      "notes": "Updated NDC (2021): 22% unconditional below BAU by 2030"
    },
    "ndc_2035": {
      "ndc3_submitted": false,
      "target_year": 2035,
      "target_pct_reduction": null,
      "reference_year": null,
      "reference_type": null,
      "ndc3_submission_date": null,
      "notes": "NDC 3.0 not yet submitted."
    },
    "net_zero": {
      "target_year": null,
      "legal_status": "none",
```

```
    "law_name": null
  },
  "cat_rating": null
},
"ZAF": {
  "country_name": "South Africa",
  "ndc_2030": {
    "target_year": 2030,
    "target_pct_reduction": 32,
    "reference_year": null,
    "reference_type": "absolute",
    "conditional": false,
    "submission_status": "updated",
    "notes": "Updated NDC (2021): 350-420 MtCO2e by 2030; ~32% below current"
  },
  "ndc_2035": {
    "ndc3_submitted": false,
    "target_year": 2035,
    "target_pct_reduction": null,
    "reference_year": null,
    "reference_type": null,
    "ndc3_submission_date": null,
    "notes": "NDC 3.0 not yet submitted."
  },
  "net_zero": {
    "target_year": 2050,
    "legal_status": "in_policy",
    "law_name": null
  },
  "cat_rating": "Highly_insufficient"
},
"MEX": {
  "country_name": "Mexico",
  "ndc_2030": {
    "target_year": 2030,
    "target_pct_reduction": 35,
    "reference_year": null,
    "reference_type": "bau",
    "conditional": false,
    "submission_status": "submitted",
    "notes": "NDC (2020): 35% unconditional GHG reduction below BAU by 2030"
  },
  "ndc_2035": {
    "ndc3_submitted": false,
    "target_year": 2035,
    "target_pct_reduction": null,
    "reference_year": null,
    "reference_type": null,
```

```
    "ndc3_submission_date": null,
    "notes": "NDC 3.0 not yet submitted."
  },
  "net_zero": {
    "target_year": 2050,
    "legal_status": "pledge",
    "law_name": null
  },
  "cat_rating": "Critically_insufficient"
},
"RUS": {
  "country_name": "Russia",
  "ndc_2030": {
    "target_year": 2030,
    "target_pct_reduction": 30,
    "reference_year": 1990,
    "reference_type": "absolute",
    "conditional": false,
    "submission_status": "updated",
    "notes": "Updated NDC (2020): 70% of 1990 levels by 2030 (30% reduction)"
  },
  "ndc_2035": {
    "ndc3_submitted": false,
    "target_year": 2035,
    "target_pct_reduction": null,
    "reference_year": null,
    "reference_type": null,
    "ndc3_submission_date": null,
    "notes": "NDC 3.0 not expected given current geopolitical situation."
  },
  "net_zero": {
    "target_year": 2060,
    "legal_status": "in_policy",
    "law_name": null
  },
  "cat_rating": "Critically_insufficient"
},
"SAU": {
  "country_name": "Saudi Arabia",
  "ndc_2030": {
    "target_year": 2030,
    "target_pct_reduction": 19,
    "reference_year": null,
    "reference_type": "bau",
    "conditional": false,
    "submission_status": "submitted",
    "notes": "NDC (2021): 278 MtCO2e annual abatement by 2030; ~19% below BAU"
  },
```

```
  "ndc_2035": {
    "ndc3_submitted": false,
    "target_year": 2035,
    "target_pct_reduction": null,
    "reference_year": null,
    "reference_type": null,
    "ndc3_submission_date": null,
    "notes": "NDC 3.0 not yet submitted."
  },
  "net_zero": {
    "target_year": 2060,
    "legal_status": "pledge",
    "law_name": null
  },
  "cat_rating": "Critically_insufficient"
},
"TUR": {
  "country_name": "Turkey",
  "ndc_2030": {
    "target_year": 2030,
    "target_pct_reduction": 41,
    "reference_year": null,
    "reference_type": "bau",
    "conditional": false,
    "submission_status": "updated",
    "notes": "Updated NDC (2023): 41% reduction below BAU by 2030"
  },
  "ndc_2035": {
    "ndc3_submitted": false,
    "target_year": 2035,
    "target_pct_reduction": null,
    "reference_year": null,
    "reference_type": null,
    "ndc3_submission_date": null,
    "notes": "NDC 3.0 not yet submitted."
  },
  "net_zero": {
    "target_year": 2053,
    "legal_status": "pledge",
    "law_name": null
  },
  "cat_rating": "Critically_insufficient"
},
"EGY": {
  "country_name": "Egypt",
  "ndc_2030": {
    "target_year": 2030,
    "target_pct_reduction": 33,
```

```
      "reference_year": null,
      "reference_type": "bau",
      "conditional": false,
      "submission_status": "updated",
      "notes": "Updated NDC (2022): 33% reduction below BAU in electricity sector by 2030"
    },
    "ndc_2035": {
      "ndc3_submitted": false,
      "target_year": 2035,
      "target_pct_reduction": null,
      "reference_year": null,
      "reference_type": null,
      "ndc3_submission_date": null,
      "notes": "NDC 3.0 not yet submitted."
    },
    "net_zero": {
      "target_year": null,
      "legal_status": "none",
      "law_name": null
    },
    "cat_rating": "Highly_insufficient"
  }
 }
}
```

파일 #6: data/supabase-bridge.md

Supabase Bridge — 기존 DB 연결 가이드

기존 Supabase 정보
- Project: VisualClimate (Climate-RAG-for-ALL 레포에서 생성)
- Tables: countries (250), indicators (67), country_data (172,121)
- Connection: .env의 SUPABASE_URL + SUPABASE_SERVICE_KEY

읽기 — 기존 데이터 가져오기

국가 메타데이터
```sql
SELECT iso3, name, region, sub_region, income_group, population
FROM countries
ORDER BY name;
```

특정 국가의 모든 지표 (최신 연도)

```
SELECT cd.indicator_code, cd.year, cd.value, cd.source, i.name, i.unit
```

```
FROM country_data cd
JOIN indicators i ON cd.indicator_code = i.code
WHERE cd.country_iso3 = '{ISO3}'
  AND cd.year = (
    SELECT MAX(year) FROM country_data
    WHERE country_iso3 = cd.country_iso3
    AND indicator_code = cd.indicator_code
  )
ORDER BY i.category, i.code;
```

시계열 (특정 국가 + 특정 지표)

```
SELECT year, value, source
FROM country_data
WHERE country_iso3 = '{ISO3}'
  AND indicator_code = '{CODE}'
ORDER BY year;
```

schema.json 지표와 Supabase 코드 매핑

schema indicator	supabase_code	변환
total_ghg_excl_lulucf	WB.EN.ATM.GHGT.KT.CE	÷1000 (kt→Mt)
co2_fossil	WB.EN.ATM.CO2E.KT	÷1000 (kt→Mt)
ghg_per_capita	WB.EN.ATM.CO2E.PC	CO2 only, not full GHG
population	WB.SP.POP.TOTL	직접 사용
gdp_ppp	WB.NY.GDP.MKTP.PP.CD	직접 사용
renewable_share	WB.EG.FEC.RNEW.ZS	Final energy, not elec
ndgain_rank	NDGAIN.RANK	직접 사용
ndgain_vulnerability	NDGAIN.VULNERABILITY	직접 사용
ndgain_readiness	NDGAIN.READINESS	직접 사용
sector_*	CTRACE.*	직접 사용

쓰기 — 새 데이터 추가

새 지표 등록 (indicators 테이블)

```
INSERT INTO indicators (source, code, name, unit, category, domain)
VALUES
```

```
  ('EDGAR', 'EDGAR.GHG.TOTAL', 'Total GHG excl. LULUCF (EDGAR)', 'MtCO2eq',
'emissions', 'emissions'),
  ('EDGAR', 'EDGAR.CH4.TOTAL', 'Total CH4 (EDGAR)', 'MtCO2eq', 'emissions',
'emissions'),
  ('IRENA', 'IRENA.RE.CAPACITY', 'Renewable energy capacity', 'GW', 'energy', 'energy'),
  ('IRENA', 'IRENA.RE.SHARE.ELEC', 'RE share of electricity', '%', 'energy', 'energy'),
  ('CAT', 'CAT.RATING', 'CAT overall rating', 'text', 'policy', 'policy'),
  ('NETZERO', 'NETZERO.YEAR', 'Net-zero target year', 'year', 'policy', 'policy'),
  ('NDC', 'NDC.3.SUBMITTED', 'NDC 3.0 submitted', 'boolean', 'policy', 'policy')
ON CONFLICT (code) DO NOTHING;
```

데이터 추가 **(country_data** 테이블**)**

```
INSERT INTO country_data (country_iso3, indicator_code, year, value, source)
VALUES ('{ISO3}', '{CODE}', {YEAR}, {VALUE}, '{SOURCE}')
ON CONFLICT DO NOTHING;
```

주의사항

- 기존 테이블 구조 절대 변경 금지
- 기존 indicator_code 절대 변경 금지
- INSERT만 허용, UPDATE/DELETE는 명시적 승인 후에만
- 새 indicator_code는 반드시 EDGAR.*, IRENA.*, CAT.*, NETZERO.*, NDC.* 패턴

파일 #9: data/tableau/tableau-setup.md

```markdown
# Tableau 연결 설계도

## 데이터 소스 연결

### 소스 1: countries_latest.csv
- 경로: data/tableau/countries_latest.csv
- 연결: Text file (CSV)
- 갱신: Claude가 /country-brief 또는 /tableau-refresh 실행 시 자동 갱신
- 역할: 모든 대시보드의 기본 데이터

### 소스 2: timeseries.csv
- 경로: data/tableau/timeseries.csv
- 연결: Text file (CSV)
- 역할: 시계열 차트용

### 관계(Relationship)
```

- countries_latest.csv ↔ timeseries.csv: ISO3 = ISO3

대시보드 3종 설계

Dashboard 1: Country Overview
- 용도: 단일 국가의 기후 현황 한눈에 보기
- 필터: 국가 선택 (Quick Filter, dropdown)
- 시트 구성:
  1. KPI 카드 4개 (총배출, 1인당, 재생에너지%, ND-GAIN 순위)
  2. 배출 추이 선 그래프 (timeseries, 최근 20년)
  3. 섹터별 파이/도넛 차트
  4. 에너지 믹스 수평 막대
  5. NDC 상태 텍스트 박스
- 크기: 1200 × 800

Dashboard 2: Country Comparison
- 용도: 2-5개국 비교
- 필터: 파라미터 5개 (Country1~5)
- 시트 구성:
  1. 비교표 (Highlight Table)
  2. 범프 차트 (지표별 순위)
  3. 레이더/거미줄 차트 (다각형 — Tableau에서는 custom polygon)
  4. 1인당 배출 수평 막대 비교
- 크기: 1400 × 900

Dashboard 3: NDC Gap Tracker
- 용도: NDC 목표 vs 실제 궤적 갭 시각화
- 필터: 국가 선택
- 시트 구성:
  1. 이중 축 선 차트: 실제 배출(실선) + 투영(점선) + 목표선(빨간 기준선)
  2. 갭 영역 (reference band or area between lines)
  3. 갭 등급 색상 KPI
  4. 연간 필요 감축률 막대
- 크기: 1200 × 700

CSV 열 규격

countries_latest.csv

ISO3,country_name,region,income_group,population,gdp_ppp,total_ghg,co2_fossil,ghg_per_capita,ghg_intensity,re_share_elec,fossil_share_pe,sector_energy_pct,sector_industry_pct,sector_agriculture_pct,sector_waste_pct,ghg_10yr_trend,ndgain_rank,ndgain_vuln,ndgain_ready,ndc30_submitted,nz_year,nz_legal,cat_rating,data_year,source_primary

timeseries.csv

ISO3,country_name,year,indicator_code,value,unit,source

Tableau 데이터 타입 매핑
CSV 열	Tableau 타입	역할
ISO3	String (Dimension)	조인 키, 필터
country_name	String (Dimension)	라벨
region	String (Dimension)	필터, 색상
income_group	String (Dimension)	필터, 색상
year	Integer (Dimension)	시간 축
population	Number (Measure)	계산
total_ghg	Number (Measure)	핵심 지표
ndc30_submitted	Boolean (Dimension)	색상/형태
cat_rating	String (Dimension)	색상

파일 #10: data/tableau/calculated-fields.md

Tableau Calculated Fields 전체 목록

파생 지표 (Calculated Fields)

CF01: GHG Per Capita

```
[total_ghg] * 1000000 / [population]
```

- 단위: tCO2eq/capita
- 용도: KPI 카드, 비교 막대, 스캐터

CF02: GHG Intensity of GDP

```
[total_ghg] * 1000000000 / [gdp_ppp]
```

- 단위: kgCO2eq/USD PPP
- 용도: 효율성 비교

CF03: GHG Trend Category

```
IF [ghg_10yr_trend] < -10 THEN “Rapid Decline”
 ELSEIF [ghg_10yr_trend] < 0 THEN “Declining”
 ELSEIF [ghg_10yr_trend] < 5 THEN “Stagnant”
 ELSEIF [ghg_10yr_trend] < 20 THEN “Rising”
 ELSE “Rapid Rise”
 END
```

- 용도: 색상 인코딩, 필터

CF04: NDC Gap (2030)

```
// timeseries에서 투영값 필요 — country_latest의 ndc_2030_target과 비교
 // 이 필드는 NDC Gap Dashboard 전용
 [projected_2030] - [ndc_2030_target]
```

CF05: NDC Gap Severity

```
IF [CF04: NDC Gap (2030)] <= 0 THEN “On Track”
 ELSEIF [CF04: NDC Gap (2030)] / [ndc_2030_target] < 0.1 THEN “Narrow Gap”
 ELSEIF [CF04: NDC Gap (2030)] / [ndc_2030_target] < 0.3 THEN “Significant Gap”
 ELSE “Critical Gap”
 END
```

- 색상: On Track=#00A67E, Narrow=#F59E0B, Significant=#E5484D, Critical=#7F1D1D

CF06: Peer Group Median (GHG Per Capita)

```
// LOD 표현식
 { FIXED [income_group], [region] : MEDIAN([CF01: GHG Per Capita]) }
```

- 용도: Country Brief의 동료 비교

CF07: Deviation from Peer Median

```
[CF01: GHG Per Capita] - [CF06: Peer Group Median (GHG Per Capita)]
```

- 양수 = 동료 그룹보다 높은 배출 (주의점)
- 음수 = 동료 그룹보다 낮은 배출 (강점)

CF08: Net Zero Countdown

```
[nz_year] - YEAR(TODAY())
```

- 용도: "X년 남음" 표시

CF09: RE Share Category

```
IF [re_share_elec] >= 80 THEN “RE Leader (80%+)”
 ELSEIF [re_share_elec] >= 50 THEN “Majority RE (50-79%)”
 ELSEIF [re_share_elec] >= 25 THEN “Growing RE (25-49%)”
```

```
 ELSE "Fossil Dominant (<25%)"
 END
```

CF10: CAT Rating Color

```
CASE [cat_rating]
 WHEN "1.5C_compatible" THEN "#00A67E"
 WHEN "Almost_sufficient" THEN "#8BC34A"
 WHEN "Insufficient" THEN "#F59E0B"
 WHEN "Highly_insufficient" THEN "#E5484D"
 WHEN "Critically_insufficient" THEN "#7F1D1D"
 ELSE "#CCCCCC"
 END
```

색상 팔레트 (brand-config.md 참조)

용도	색상
Changer	#00A67E
Starter	#F59E0B
Talker	#E5484D
Primary	#0066FF
On Track	#00A67E
Narrow Gap	#F59E0B
Significant Gap	#E5484D
Critical Gap	#7F1D1D
Data Missing	#CCCCCC

대시보드별 사용 필드

Dashboard 1: Country Overview
- CF01 (GHG/capita), CF03 (Trend Category), CF08 (NZ Countdown), CF09 (RE Category)

Dashboard 2: Country Comparison
- CF01, CF02, CF06, CF07, CF09, CF10

Dashboard 3: NDC Gap Tracker
- CF04, CF05, CF08

파일 **#11~#20: Claude** 스킬 **10**개

파일 **#11: .claude/skills/schema-explainer/SKILL.md**

---

name: schema-explainer
description: Analyzes new datasets and maps columns to our schema.json indicators. Use when onboarding a new data source or when a source updates its column names.

Schema Explainer

Trigger
/explain-schema

What This Does
Takes raw CSV column names or first 20 rows, maps them to data/schema.json indicators, flags unmapped columns, detects unit mismatches, and reports missing coverage.

Input
User pastes: CSV header row, OR first 20 rows, OR column name list

Process
1. Read data/schema.json to load all 23 indicator definitions
2. For each input column:
   a. Fuzzy-match column name to schema indicator labels
   b. Check value range to infer unit (e.g., 0.1-50 → tCO2eq/capita likely)
   c. If matched: confirm unit compatibility
   d. If not matched: flag as "not in schema — review for addition"
3. Check coverage: which schema indicators have NO matching column
4. Detect missingness pattern: which countries/years have gaps

Output Format
Column Mapping
#	Source Column	Matched Indicator	Schema Code	Unit Match	Missing %	Notes
1	{col}	{indicator or "NO MATCH"}	{code}	✅/⚠️	{%}	{notes}

Coverage Gap
Schema indicators with no matching source column:
- {indicator_1}: Consider source {X}
- {indicator_2}: Not available in this dataset

Recommended Actions
1. {action 1}
2. {action 2}

Rules
- Always read data/schema.json first — never guess indicator definitions
- Unit mismatch = ⚠️, not auto-convert. Report and let user decide.
- Never add columns to schema without explicit approval

파일 #12: .claude/skills/source-comparator/SKILL.md

---
name: source-comparator
description: Compares 2-4 climate data sources on a topic using source-registry.json metadata. Produces comparison table + practical recommendations.

Source Comparator

Trigger
/compare-sources {topic}

What This Does
For a given topic (e.g., "GHG emissions", "renewable energy", "vulnerability"),
pulls relevant sources from data/source-registry.json and produces a structured comparison.

Input
Topic keyword: "GHG emissions", "energy", "vulnerability", "NDC policy", etc.

Process
1. Load data/source-registry.json
2. Filter sources relevant to topic
3. For each source, extract these 17 fields:
   full_name, publisher, url_data, underlying_sources (if secondary),
   methodology, gwp, gases, sectors, coverage_countries, coverage_years,
   time_resolution, spatial_resolution, update_frequency, format,
   license, strengths, limitations
4. Generate comparison table (sources × key fields)
5. Explain WHY numbers differ between sources (methodology, scope, GWP)
6. Recommend: which source for which use case

Output Format
Source Cards
{Source 1 Name}
Field	Detail
Publisher	{publisher}
URL	{url_data}
... (all 17 fields)

Comparison Table
Field	{Source1}	{Source2}	{Source3}
Countries	...	...	...
... (key differentiators)

Why Numbers Differ

{1 paragraph explaining methodology/scope/GWP differences}

Practical Recommendation
- For NDC reporting: use {X} because...
- For ESG disclosure: use {Y} because...
- For academic research: use {Z} because...

Rules
- ONLY use sources from data/source-registry.json — never invent sources
- All URLs must be direct data download pages, not homepages
- If a field is unknown, mark as "⚠️ Verify at {url}"

파일 #13: .claude/skills/data-quality-checker/SKILL.md

---
name: data-quality-checker
description: Diagnoses data quality issues using IPCC TCCCA principles (Transparency, Completeness, Consistency, Comparability, Accuracy). Produces issue table + fix recommendations.

Data Quality Checker

Trigger
/check-quality

What This Does
Receives raw data (CSV paste or file reference) and checks for 7 categories of issues using IPCC quality principles.

Input
User pastes data or references a file path.

Process
1. Unit consistency: same column mixing kt/Mt/Gg?
2. GWP consistency: AR4/AR5/AR6 values mixed?
3. Missing data: by country (rows) and by year (columns)
4. Time series continuity: year-over-year change > ±30% → anomaly
5. Sector classification: IPCC 1996 vs 2006 categories mixed?
6. LULUCF mixed: some rows include, some exclude?
7. Double counting: international bunkers, indirect CO2?

Output Format
Issue Table
#	Issue Type	Location	Severity	TCCCA Principle	Fix
1	{type}	{col/row}	🔴/🟡/🟢	{principle}	{fix}

Summary
- **Overall Verdict**: {PASS / WARNING / FAIL}
- **Usable rows**: {X} of {Y} ({%})
- **Usable countries**: {X} of {Y}
- **Recommended next step**: {action}

Severity Scale
- 🔴 Critical: Makes data unusable for comparison (mixed LULUCF, wrong GWP)
- 🟡 Warning: Reducible with documented assumptions (missing years, unit conversion)
- 🟢 Note: Best practice improvement (rounding, label standardization)

Rules
- Never auto-fix data — only diagnose and recommend
- Reference IPCC TCCCA explicitly for each issue
- If data has < 70% completeness for a country, recommend excluding it

파일 #14: .claude/skills/indicator-definer/SKILL.md

---
name: indicator-definer
description: Defines climate/ESG indicators using IPCC/UNFCCC/GHG Protocol official definitions. Produces term cards + comparison table + usage checklist.

Indicator Definer

Trigger
/define {term group}

What This Does
Takes a group of related or confused terms (e.g., "GHG Intensity vs Carbon Intensity vs Emission Factor") and produces precise definitions, comparisons, and usage checklist.

Input
Term group as quoted string.

Process
1. For each term:
   a. Official definition (IPCC/UNFCCC/GHG Protocol source)
   b. One-line plain English explanation
   c. Unit
   d. Use context (when/where this term appears)
   e. Common confusion point
   f. Real numeric example with source+year
2. Comparison table across all terms
3. Usage checklist: what to verify when using these terms in a report

Output Format
Term Card: {Term 1}
Field	Detail
Official Name	{name}
Definition	{IPCC/UNFCCC definition}
Source	{reference}
Unit	{unit}
Use Context	{context}
⚠️ Confusion	{what people mix up}
Example	{country}: {value} {unit} ({source}, {year})

Comparison Table
Term	Definition (short)	Unit	Typical Use	Watch Out

Usage Checklist
When writing a report with these terms:
☐ GWP basis stated (AR4/AR5/AR6)
☐ Scope boundary clear (Scope 1/2/3, national/territorial)
☐ Base year explicitly mentioned
☐ LULUCF inclusion/exclusion stated
☐ Source + vintage year cited for every number

Rules
- All definitions must cite IPCC/UNFCCC/GHG Protocol — no Wikipedia-level definitions
- Examples must use real numbers from data/ or Supabase — never fabricate
- If a term has different meanings in different frameworks, list ALL meanings

파일 #15: .claude/skills/country-brief/SKILL.md

---
name: country-brief
description: Generates a 1-page Country Climate Brief for any ISO3 country. Reads from Supabase + data/ files. Follows templates/country-brief.md format.

Country Brief Generator

Trigger
/country-brief {ISO3}

What This Does
Generates a structured Country Climate Brief — the CORE PRODUCT of this engine. Covers emissions, energy, policy, vulnerability for a single country.

Data Loading Sequence
1. Read data/schema.json — know what 23 indicators to look for
2. Query Supabase: latest values for this ISO3 (using supabase_code mappings)
3. Read data/ndc-targets-v2.json — NDC 2030 + 2035 + net zero
4. Read data/source-registry.json — for accurate source citations
5. If country is in existing JSON files:
   - public/data/kaya/{ISO3}.json — Kaya decomposition
   - public/data/ndc-gap/{ISO3}.json — NDC gap analysis
   - data/climate-trace-ghg.json — sector breakdown (6 countries only)
   - data/ndgain-scores.json — vulnerability scores

Process
1. Load all available data for {ISO3}
2. Verify each indicator: has source? has year? → missing = [DATA MISSING]
3. Unit conversions per schema.json (e.g., WB kt → Mt: ÷1000)
4. Calculate derived indicators:
   - ghg_per_capita = total_ghg * 1e6 / population
   - ghg_intensity = total_ghg * 1e9 / gdp_ppp
   - 10yr_trend = ((latest / 10yr_prior) - 1) × 100
5. Identify peer group:
   - Query Supabase: same income_group + same region
   - Pick top 3 by population (excluding this country)
   - Get their indicator values
6. Auto-derive strengths (better than peer median) and watchpoints (worse)
7. Write Executive Summary: 4 sentences, 80 words max
8. Fill templates/country-brief.md template
9. Auto-update data/tableau/countries_latest.csv with this country's row

Output
Save to: output/{ISO3}-country-brief-{YYYY-MM-DD}.md
Format: templates/country-brief.md

Rules
- NO invented numbers. If data missing → [DATA MISSING]
- Cross-check: if EDGAR and Climate TRACE differ > 10% → ⚠️ + footnote
- All comparisons: same source, same GWP, same year
- Peer comparison: use Supabase income_group + region from countries table
- Update Tableau CSV after every brief generation

파일 **#16: .claude/skills/comparison-brief/SKILL.md**

---
name: comparison-brief
description: Generates a Country Comparison Brief for 2-5 countries. 8 core indicators, top 3 differences, common challenges.

Comparison Brief Generator

Trigger
/compare-countries {ISO3_1} {ISO3_2} [ISO3_3] [ISO3_4] [ISO3_5]

What This Does
Compares 2-5 countries across 8 core indicators. Identifies key differences
and common challenges. Follows templates/comparison-brief.md format.

Data Loading
For each ISO3: same loading sequence as country-brief (Supabase + data/ files)

Process
1. Load data for all countries
2. Auto-detect comparison type:
   - Same region → "Regional Comparison"
   - Same income_group → "Peer Comparison"
   - Mixed → "Cross-Category Comparison"
3. Build comparison table: 8 indicators × N countries
   Indicators: total_ghg, ghg_per_capita, ghg_intensity, re_share_elec,
   ndc30_submitted, nz_year, ndgain_rank, ghg_10yr_trend
4. Normalize each indicator to z-scores
5. Find top 3 largest differences (highest z-score spread)
6. Find common challenges: indicators where ALL countries are worse than global median
7. Write insight paragraph (100 words max)

Output
Save to: output/compare-{ISO3s joined by -}-{YYYY-MM-DD}.md
Format: templates/comparison-brief.md

Rules
- Minimum 2, maximum 5 countries
- If any country has [DATA MISSING] for an indicator, exclude that indicator from comparison
- All values same source + same year + same GWP

파일 #17: .claude/skills/ndc-gap-brief/SKILL.md

---
name: ndc-gap-brief
description: Quantitative NDC gap analysis for a single country. Compares NDC targets vs emission trajectory. Uses data/ndc-targets-v2.json.

NDC Gap Brief Generator

Trigger
/ndc-gap {ISO3}

What This Does
Calculates the gap between a country's NDC target and its actual/projected emissions. Assigns a gap severity rating. Follows templates/ndc-gap-brief.md format.

Data Loading
1. data/ndc-targets-v2.json → NDC 2030/2035 targets for this country
2. Supabase → historical emissions timeseries (WB.EN.ATM.GHGT.KT.CE, convert to Mt)
3. If exists: public/data/ndc-gap/{ISO3}.json → existing gap analysis
4. data/schema.json → unit/conversion rules

Process
1. Load NDC target:
   - Determine target type: absolute / intensity / BAU%
   - Get reference year, reference emissions, target emissions
   - Distinguish conditional vs unconditional
2. Load emission timeseries (at least 5 recent years)
3. Calculate 5-year linear trend (regression)
4. Project to 2030 and 2035
5. Calculate gap:
   - For absolute targets: gap = projected - target
   - For intensity targets: convert using GDP projection (or note [GDP PROJECTION NEEDED])
   - For BAU targets: note that BAU baseline uncertainty makes gap calculation approximate
6. Assign gap rating:
   - On Track: projected ≤ target
   - Narrow Gap (<10%): policy acceleration can close
   - Significant Gap (10-30%): major policy shift needed
   - Critical Gap (>30%): structural transformation needed
7. Calculate required annual reduction rate to meet target
8. Identify 3 risks/watchpoints

Output
Save to: output/{ISO3}-ndc-gap-{YYYY-MM-DD}.md
Format: templates/ndc-gap-brief.md

Rules
- For intensity/BAU targets, clearly state assumption limitations
- Use LOWER bound of NDC ranges (conservative)
- If < 5 years of emission data available, mark analysis as "LOW CONFIDENCE"
- Always include both 2030 and 2035 analysis (2035 may be [NDC 3.0 NOT SUBMITTED])

파일 **#18: .claude/skills/qa-checker/SKILL.md**

---
name: qa-checker

description: Quality assurance for generated reports. Checks factual accuracy, unit consistency, source citations, GWP consistency, temporal alignment.

QA Checker

Trigger
/qa {file_path}

What This Does
Reads a generated report or content file and validates against source data.

Process
1. **Source Citation Check**: Every number must have (Source, Year)
   - Count numbers without citations → ⚠️ for each
2. **Factual Cross-Check**: Compare stated numbers against:
   - data/ndc-targets-v2.json
   - data/source-registry.json
   - Supabase (query for ISO3 + indicator)
   - data/*.json files
   - Tolerance: ≤5% difference = ✅, >5% = ⚠️
3. **Unit Consistency**: Within the same report:
   - Same indicator should use same unit throughout
   - MtCO2eq vs ktCO2eq mixing = 🔴
4. **GWP Consistency**: All values should be AR5
   - Any AR4 or AR6 reference without conversion note = 🔴
5. **Temporal Alignment**: Comparison values should be same year
   - Year difference > 2 between compared values = ⚠️
6. **LULUCF Consistency**: No mixing incl/excl in same table
7. **Completeness**: Check for [DATA MISSING] tags — count them

Output Format
QA Report for: {filename}

#	Check	Status	Detail
1	Citations	✅/⚠️	{X} numbers, {Y} cited, {Z} uncited
2	Factual: {claim}	✅/⚠️	Reference: {ref_value}, Stated: {stated}, Diff: {%}
3	Units	✅/⚠️/🔴	{detail}
4	GWP	✅/⚠️/🔴	{detail}
5	Years	✅/⚠️	Max year gap: {N}
6	LULUCF	✅/🔴	{detail}
7	Completeness	✅/⚠️	{N} [DATA MISSING] tags

Verdict: {PASS / PASS WITH WARNINGS / FAIL}

Required Fixes
1. {fix 1}

2. {fix 2}

Rules
- Run this after EVERY report generation — no exceptions
- FAIL = report cannot be published until fixes applied
- PASS WITH WARNINGS = publishable but note the warnings

파일 #19: .claude/skills/content-repurposer/SKILL.md

---
name: content-repurposer
description: Transforms reports into different content formats — LinkedIn carousel, text post, executive summary, teaching slides, email newsletter.

Content Repurposer

Trigger
/repurpose {file_path} {format}

Formats
- linkedin-carousel
- linkedin-text
- executive-summary
- teaching-slide
- email-newsletter

What This Does
Reads a generated report and restructures it for a specific distribution format.
All numbers keep their sources. No new data added.

Process
1. Read source report
2. Extract 3 core messages (most surprising/impactful findings)
3. Reshape per format rules below
4. Maintain all (Source, Year) citations
5. Add format-appropriate CTA

Format: linkedin-carousel
- Slide 1: Hook (≤15 words, includes a number, addresses practitioner pain point)
- Slides 2-4: Core information (≤80 chars per text block, tables ≤4×5)
- Last Slide: CTA + 1-sentence summary
- Caption: 150-250 chars + 5 hashtags max
- Save to: output/{source-name}-carousel-{YYYY-MM-DD}.md

Format: linkedin-text
- 1,300 chars max

- Structure: Hook → Data Point 1 → Data Point 2 → So What → CTA
- No slides, pure text
- Save to: output/{source-name}-linkedin-text-{YYYY-MM-DD}.md

Format: executive-summary
- 3 sentences only
- Sentence 1: Current state (number)
- Sentence 2: Key finding (comparison or trend)
- Sentence 3: Implication (what this means for practitioners)
- Save to: output/{source-name}-exec-summary-{YYYY-MM-DD}.md

Format: teaching-slide
- 10 slides outline
- Slide 1: Title + learning objective
- Slides 2-8: Content (1 concept per slide, 1 visual suggestion per slide)
- Slide 9: Summary
- Slide 10: Discussion questions (3)
- Save to: output/{source-name}-teaching-slides-{YYYY-MM-DD}.md

Format: email-newsletter
- Subject line (≤60 chars, number + urgency)
- Preview text (≤90 chars)
- Body: Hook → 3 Key Findings → CTA button text
- ≤500 words total
- Save to: output/{source-name}-newsletter-{YYYY-MM-DD}.md

Rules
- NEVER add numbers not in the source report
- Every number must keep its (Source, Year) citation
- LinkedIn hashtags: #ClimateData #ESG #NDC #NetZero #Sustainability (select relevant 5)

파일 #20: .claude/skills/tableau-spec-writer/SKILL.md

---
name: tableau-spec-writer
description: Generates Tableau dashboard specifications including data source connections, sheet designs, calculated fields, filter logic, and layout. Produces a blueprint that Harim can directly implement in Tableau.

Tableau Spec Writer

Trigger
/tableau-spec {dashboard_type}

Dashboard Types
- country-overview

- country-comparison
- ndc-gap-tracker
- custom (user describes)

What This Does
Produces a complete Tableau implementation blueprint: data sources, sheets, calculated fields, filters, parameters, layout, colors, tooltips.
Harim takes this spec and builds it directly in Tableau Desktop — no guessing.

Process
1. Read data/tableau/tableau-setup.md for CSV structure
2. Read data/tableau/calculated-fields.md for existing calc fields
3. Read brand-config.md for colors/fonts
4. Generate spec per dashboard type

Output Format
Tableau Dashboard Spec: {Dashboard Name}

Data Sources
Source	File	Join/Relationship
{name}	{path}	{join key}

Parameters
Parameter	Type	Default	Allowable Values
{name}	{type}	{default}	{values}

Calculated Fields (New)
Name	Formula	Purpose
{name}	{Tableau formula}	{what it does}

Sheets
Sheet 1: {Name}
- Chart Type: {bar/line/scatter/map/etc.}
- Rows: {field}
- Columns: {field}
- Color: {field} → {palette}
- Size: {field or fixed}
- Label: {field}
- Tooltip: {custom tooltip text with fields}
- Filter: {field}
- Sort: {field, direction}

Sheet 2: {Name}
... (repeat)

Dashboard Layout

```
┌──────────────────────────────────────────────┐
│ Title Bar: {text} │
├──────────────┬──────────────────────────────┤
│ Filter │ Sheet 1 (w×h) │
│ Panel ├──────────────────────────────┤
│ │ Sheet 2 (w×h) │
├──────────────┴──────────────┬───────────────┤
│ Sheet 3 (w×h) │ Sheet 4 (w×h) │
└──────────────────────────────┴───────────────┘
```

- Size: {width} × {height}
- Device: Desktop

Formatting
- Background: #FFFFFF
- Title font: Inter Bold 14pt #1A1A2E
- Axis font: Inter Regular 10pt #4A4A6A
- Gridlines: Light gray #E5E7EB, dashed
- Borders: None (clean look)

Interactivity
- Filter actions: {source sheet} → {target sheet} on {field}
- Highlight actions: {description}
- URL actions: {if any}

Rules
- Always reference existing calculated-fields.md — don't reinvent
- Colors must match brand-config.md
- Every tooltip must include (Source, Year) for data credibility
- Dashboard size must be specified exactly
- Recommend LOD expressions where needed for peer comparison

파일 #21: .claude/agents/engine-director.md

---
name: engine-director
description: Master orchestrator for the Climate Intelligence Engine. Routes tasks to appropriate skills, manages workflow sequence, enforces quality gates.
tools: Bash, Read, Write, Edit, Grep, Glob
model: opus
skills:
  - schema-explainer
  - source-comparator

```
  - data-quality-checker
  - indicator-definer
  - country-brief
  - comparison-brief
  - ndc-gap-brief
  - qa-checker
  - content-repurposer
  - tableau-spec-writer
---

> **ALWAYS read CLAUDE.md first before any task.**

# Engine Director

## Role
You are the master orchestrator for the VisualClimate Trojan engine.
You coordinate all skills, enforce data rules, and ensure quality gates.

## Workflow: Country Brief Pipeline
1. /country-brief {ISO3}
   → skill: country-brief → output/{ISO3}-country-brief-{date}.md
2. /qa output/{ISO3}-country-brief-{date}.md
   → skill: qa-checker → PASS required before next step
3. /repurpose output/{ISO3}-country-brief-{date}.md linkedin-carousel
   → skill: content-repurposer → output/{ISO3}-carousel-{date}.md
4. /tableau-spec country-overview
   → skill: tableau-spec-writer → output/tableau-country-overview-spec.md

## Workflow: Comparison Pipeline
1. /compare-countries {ISO3s}
   → skill: comparison-brief
2. /qa {output}
3. /repurpose {output} linkedin-carousel
4. /tableau-spec country-comparison

## Workflow: NDC Gap Pipeline
1. /ndc-gap {ISO3}
   → skill: ndc-gap-brief
2. /qa {output}
3. /repurpose {output} linkedin-carousel
4. /tableau-spec ndc-gap-tracker

## Workflow: Data Onboarding
1. /explain-schema (paste new CSV)
   → skill: schema-explainer
2. /check-quality (paste data)
   → skill: data-quality-checker → must PASS
3. Run scripts/push-to-supabase.ts for clean data
```

Quality Gates (Non-Negotiable)
- Every report MUST pass /qa before distribution
- Every number MUST have (Source, Year)
- GWP must be AR5 across entire output
- LULUCF must be consistent within each report
- [DATA MISSING] is acceptable; invented data is not.

Skill Quick Reference
Command	Skill	Output
/explain-schema	schema-explainer	Column mapping table
/compare-sources {topic}	source-comparator	Source comparison + recommendation
/check-quality	data-quality-checker	Issue table + verdict
/define {terms}	indicator-definer	Term cards + comparison + checklist
/country-brief {ISO3}	country-brief	Country Climate Brief (MD)
/compare-countries {ISO3s}	comparison-brief	Comparison Brief (MD)
/ndc-gap {ISO3}	ndc-gap-brief	NDC Gap Brief (MD)
/qa {path}	qa-checker	QA report + verdict
/repurpose {path} {fmt}	content-repurposer	Reformatted content
/tableau-spec {type}	tableau-spec-writer	Tableau implementation blueprint

Data Access Priority
1. Supabase (172,121 rows) — primary
2. data/*.json in this repo — supplementary
3. data/ in Climate-RAG-for-ALL repo — reference
4. External API (live fetch) — only if explicitly requested

파일 **#22~#26:** 템플릿

파일 **#22: templates/country-brief.md**

Country Climate Brief: {COUNTRY_NAME} ({ISO3})

Data vintage: {LATEST_DATA_YEAR} | **Generated**: {TODAY_DATE} | **Engine**: VisualClimate Trojan v1.0

Executive Summary
{4 sentences, 80 words max. Sentence 1: emissions status. Sentence 2: energy transition. Sentence 3: policy alignment. Sentence 4: overall assessment.}

Key Metrics

Indicator	Value	Unit	Source	Year
Total GHG (excl. LULUCF)	{value}	$MtCO_2eq$	{source}	{year}
CO_2 from fossil fuels	{value}	$MtCO_2$	{source}	{year}
GHG per capita	{value}	tCO_2eq/cap	derived	{year}
GHG intensity of GDP	{value}	$kgCO_2eq$/USD PPP	derived	{year}
Renewable share (electricity)	{value}	%	{source}	{year}
10-year GHG trend	{value}	%	derived	{base_yr}-{latest_yr}
ND-GAIN rank	{value}	rank/181	ND-GAIN	{year}

Emissions Profile
{Paragraph: sector shares (energy, industry, agriculture, waste), gas composition, 10-year trend narrative. Numbers with sources.}

Energy Transition
{Paragraph: current energy mix, renewable share trajectory, coal status, fossil fuel dependency. Numbers with sources.}

Policy Alignment

Item	Status
NDC 3.0 submitted	{Yes/No}
2030 target	{description + number}
2035 target	{description + number, or "NDC 3.0 not submitted"}
Net-zero target	{year}, {legal_status}
CAT rating	{rating, or "Not rated by CAT"}

Peer Comparison
Peer group: {income_group} × {region} (top 3 by population)

Indicator	{THIS_COUNTRY}	{PEER_1}	{PEER_2}	{PEER_3}	Peer Median
GHG/capita (tCO_2eq)	{v}	{v}	{v}	{v}	{v}
RE share (%)	{v}	{v}	{v}	{v}	{v}
GHG intensity	{v}	{v}	{v}	{v}	{v}
ND-GAIN rank	{v}	{v}	{v}	{v}	{v}

Strengths & Watchpoints
Strengths (better than peer median):
1. {strength 1 — indicator, value, comparison}
2. {strength 2}
3. {strength 3}

Watchpoints (worse than peer median):
1. {watchpoint 1 — indicator, value, comparison}
2. {watchpoint 2}
3. {watchpoint 3}

Sources: {list all sources with URLs}
Methodology: GWP=AR5 (IPCC 2013), LULUCF excluded, peer group={income_group}+{region}
Disclaimer: This brief uses publicly available data and automated analysis. Verify critical decisions with primary sources.

파일 #23: templates/comparison-brief.md

Country Comparison Brief: {COUNTRY_1} vs {COUNTRY_2} [vs ...]

Comparison type: {Regional / Peer / Cross-Category} | **Generated**: {TODAY_DATE}

Comparison Table

Indicator	{C1}	{C2}	{C3?}	{C4?}	{C5?}	Unit	Source
Total GHG	{v}	{v}	{v}	{v}	{v}	$MtCO_2eq$	{src}
GHG/capita	{v}	{v}	{v}	{v}	{v}	tCO_2eq	derived
GHG intensity	{v}	{v}	{v}	{v}	{v}	$kgCO_2eq/USD$	derived
RE share (elec)	{v}	{v}	{v}	{v}	{v}	%	{src}
NDC 3.0 submitted	{v}	{v}	{v}	{v}	{v}	Y/N	UNFCCC
Net-zero year	{v}	{v}	{v}	{v}	{v}	year	NZ Tracker
ND-GAIN rank	{v}	{v}	{v}	{v}	{v}	rank	ND-GAIN
10yr GHG trend	{v}	{v}	{v}	{v}	{v}	%	derived

Top 3 Differences
1. **{Indicator}**: {C_high} at {value} vs {C_low} at {value} — {2-sentence explanation}
2. **{Indicator}**: {explanation}
3. **{Indicator}**: {explanation}

Common Challenges
{Paragraph: indicators where ALL countries underperform global median}

Implications
{100 words: what these differences mean for practitioners}

Sources: {list}
All values: same source, same GWP (AR5), same year where possible

파일 #24: templates/ndc-gap-brief.md

NDC Progress & Gap Brief: {COUNTRY_NAME} ({ISO3})

NDC 3.0 submitted: {Yes/No} | **Generated**: {TODAY_DATE}

NDC Targets Summary

Item	Value
Reference year	{year}
Reference emissions	{value} $MtCO_2eq$
2030 target	{description + value}
2035 target	{description + value, or "NDC 3.0 not submitted"}
Target type	{absolute / intensity / BAU %}
Conditional elements	{yes/no + description}

Actual Trajectory

- Latest emissions ({year}): {value} $MtCO_2eq$ ({source})
- 5-year trend: {+/-X}% annually
- Projected 2030: {value} $MtCO_2eq$
- Projected 2035: {value} $MtCO_2eq$
- Projection method: Linear regression on {start_year}-{end_year}

Gap Analysis

Metric	2030	2035
Target	{value} $MtCO_2eq$	{value} $MtCO_2eq$
Projected	{value} $MtCO_2eq$	{value} $MtCO_2eq$
Gap	{value} $MtCO_2eq$ ({%})	{value} $MtCO_2eq$ ({%})
Required annual reduction	{%}/yr	{%}/yr

Gap Rating: {On Track / Narrow Gap / Significant Gap / Critical Gap}

{1-sentence explanation of what this rating means}

Risks & Watchpoints
1. {Risk 1 — specific, with number if possible}

2. {Risk 2}
3. {Risk 3}

Sources: {list}
Methodology: Linear projection from {N}-year trend. GWP=AR5. LULUCF excluded.
Limitation: {target_type}-based targets require additional assumptions — see notes.

파일 #25: templates/linkedin-carousel.md

LinkedIn Carousel: {TITLE}

Source report: {source_file_path}
Generated: {TODAY_DATE}

Slide 1 — Hook
{≤15 words. Must include a number. Must address a practitioner pain point.}

Slide 2 — {Subtitle}
{≤80 characters main text}
{Optional: table ≤4 columns × 5 rows}

Slide 3 — {Subtitle}
{≤80 characters main text}
{Optional: single key comparison or chart description}

Slide 4 — {Subtitle}
{≤80 characters main text}
{Optional: "So what?" implication}

Slide 5 — CTA
{1-sentence summary of the entire carousel}
{CTA: "Follow for weekly climate data insights" or "DM for full brief"}

Caption
{150-250 characters}

{Hashtags: #ClimateData #ESG #NDC #NetZero #Sustainability}

Design Notes for Canva/Figma

- Size: 1080 × 1080 px
- Background: #FFFFFF
- Primary color: #0066FF
- Accent: {#00A67E for positive, #E5484D for negative}
- Font: Inter (heading Bold 24pt, body Regular 16pt)
- Every slide: VisualClimate logo bottom-right, source citation bottom-left

파일 #26: templates/tableau-dashboard-spec.md

Tableau Dashboard Spec: {DASHBOARD_NAME}

Generated: {TODAY_DATE} | **For**: Harim (Tableau Desktop)

Data Sources

#	Source Name	File Path	Join Key
1	{name}	{path}	{key}

Parameters

Parameter	Data Type	Current Value	Allowable Values
{name}	{String/Integer/Float}	{default}	{list or range}

Calculated Fields

#	Name	Formula	Return Type	Purpose
1	{name}	{Tableau formula}	{type}	{purpose}

Sheets

Sheet 1: {Name}
- **Chart Type**: {type}
- **Rows**: {field(s)}
- **Columns**: {field(s)}
- **Color**: {field} using {palette}
- **Size**: {field or fixed}
- **Label**: {field(s)}
- **Tooltip**:

{Country}: {value} {unit}
 Source: {source} ({year})

- **Filter**: {field}
- **Sort**: {field} {ASC/DESC}
- **Reference Line**: {if any}

Sheet 2: {Name}
{...repeat structure...}

Dashboard Layout

```
┌──────────────────────────────────────────────────────────────┐
│ Title: {title text} │
├──────────────────┬──────────────────────────────────────────┤
│ Filters │ Sheet 1 │
│ │ ({width} × {height}) │
│ ├──────────────────────────────────────────┤
│ │ Sheet 2 │
│ │ ({width} × {height}) │
├──────────────────┴──────────────────────┬──────────────────────┤
│ Sheet 3 │ Sheet 4 │
│ ({width} × {height}) │ ({w} × {h}) │
└──────────────────────────────────────────┴──────────────────────┘
```

- **Total Size**: {width} × {height}
- **Device Layout**: Desktop (primary), Tablet (secondary)

Formatting
- Background: #FFFFFF
- Title: Inter Bold 14pt #1A1A2E
- Subtitle: Inter Regular 11pt #4A4A6A
- Axis labels: Inter Regular 10pt #4A4A6A
- Gridlines: #E5E7EB dashed
- Borders: None

Interactivity
- **Filter Actions**: {source sheet} → {target sheets} on {field}
- **Highlight Actions**: {description}
- **URL Actions**: {if any}

Color Encoding
Value/Category	Color
{value}	{hex}

Reference: data/tableau/calculated-fields.md, brand-config.md

---

파일 #28: scripts/pull-from-supabase.ts

```
/**
 * pull-from-supabase.ts
 *
 * Pulls data from the existing VisualClimate Supabase database
 * and writes it to local data/ files for the Trojan engine.
 *
 * Usage: npx tsx scripts/pull-from-supabase.ts [ISO3]
 *   - No args: pulls all 250 countries (latest values only)
 *   - With ISO3: pulls single country (full timeseries)
 *
 * Output:
 *   - data/tableau/countries_latest.csv (all countries, latest year per indicator)
 *   - data/tableau/timeseries.csv (if ISO3 specified, full timeseries)
 */

import { createClient } from '@supabase/supabase-js';
import * as fs from 'fs';
import * as path from 'path';

const SUPABASE_URL = process.env.SUPABASE_URL!;
const SUPABASE_KEY = process.env.SUPABASE_SERVICE_KEY!;

const supabase = createClient(SUPABASE_URL, SUPABASE_KEY);

// Schema indicator → Supabase code mapping (from data/schema.json)
const SCHEMA_MAP: Record<string, { code: string; conversion?: string }> = {
  'total_ghg': { code: 'WB.EN.ATM.GHGT.KT.CE', conversion: 'divide_1000' },
  'co2_fossil': { code: 'WB.EN.ATM.CO2E.KT', conversion: 'divide_1000' },
  'ghg_per_capita_co2_only': { code: 'WB.EN.ATM.CO2E.PC' },
  'population': { code: 'WB.SP.POP.TOTL' },
  'gdp_ppp': { code: 'WB.NY.GDP.MKTP.PP.CD' },
  're_share_final_energy': { code: 'WB.EG.FEC.RNEW.ZS' },
  'ndgain_rank': { code: 'NDGAIN.RANK' },
  'ndgain_vulnerability': { code: 'NDGAIN.VULNERABILITY' },
  'ndgain_readiness': { code: 'NDGAIN.READINESS' },
  'ctrace_power': { code: 'CTRACE.POWER' },
  'ctrace_transportation': { code: 'CTRACE.TRANSPORTATION' },
  'ctrace_manufacturing': { code: 'CTRACE.MANUFACTURING' },
  'ctrace_agriculture': { code: 'CTRACE.AGRICULTURE' },
  'ctrace_waste': { code: 'CTRACE.WASTE' },
  'ember_carbon_intensity': { code: 'EMBER.CARBON.INTENSITY' },
};
```

```
function applyConversion(value: number, conversion?: string): number {
  if (conversion === 'divide_1000') return value / 1000;
  return value;
}

async function pullCountriesLatest() {
  console.log('Pulling latest values for all 250 countries...');

  // Get all countries
  const { data: countries } = await supabase
    .from('countries')
    .select('iso3, name, region, income_group, population');

  if (!countries) { console.error('No countries found'); return; }

  const rows: string[] = [];
  const header =
'ISO3,country_name,region,income_group,population,gdp_ppp,total_ghg,co2_fossil,ghg_per
_capita,re_share_elec,ndgain_rank,ndgain_vuln,ndgain_ready,data_year,source_primary';
  rows.push(header);

  for (const country of countries) {
    // Get latest year for each key indicator
    const { data: latestData } = await supabase
      .from('country_data')
      .select('indicator_code, year, value, source')
      .eq('country_iso3', country.iso3)
      .in('indicator_code', Object.values(SCHEMA_MAP).map(m => m.code))
      .order('year', { ascending: false });

    if (!latestData || latestData.length === 0) continue;

    // Get latest value per indicator
    const latest: Record<string, { value: number; year: number; source: string }> = {};
    for (const row of latestData) {
      if (!latest[row.indicator_code]) {
        latest[row.indicator_code] = { value: row.value, year: row.year, source: row.source };
      }
    }

    const get = (schemaKey: string): string => {
      const mapping = SCHEMA_MAP[schemaKey];
      if (!mapping) return '';
      const d = latest[mapping.code];
      if (!d) return '';
      return String(applyConversion(d.value, mapping.conversion));
    };
```

```
    const getYear = (): string => {
      const years = Object.values(latest).map(d => d.year);
      return years.length > 0 ? String(Math.max(...years)) : '';
    };

    // Calculate derived: ghg_per_capita
    const totalGhg = get('total_ghg');
    const pop = get('population');
    const ghgPerCapita = totalGhg && pop && Number(pop) > 0
      ? String((Number(totalGhg) * 1000000) / Number(pop))
      : '';

    const row = [
      country.iso3,
      `"${country.name}"`,
      `"${country.region || ''}"`,
      `"${country.income_group || ''}"`,
      get('population'),
      get('gdp_ppp'),
      get('total_ghg'),
      get('co2_fossil'),
      ghgPerCapita,
      get('re_share_final_energy'),
      get('ndgain_rank'),
      get('ndgain_vulnerability'),
      get('ndgain_readiness'),
      getYear(),
      'Supabase'
    ].join(',');

    rows.push(row);
  }

  const outPath = path.join(__dirname, '..', 'data', 'tableau', 'countries_latest.csv');
  fs.mkdirSync(path.dirname(outPath), { recursive: true });
  fs.writeFileSync(outPath, rows.join('\n'), 'utf-8');
  console.log(`Written ${rows.length - 1} countries to ${outPath}`);
}

async function pullTimeseries(iso3: string) {
  console.log(`Pulling timeseries for ${iso3}...`);

  const { data: country } = await supabase
    .from('countries')
    .select('name')
    .eq('iso3', iso3)
    .single();
```

```
  const { data: tsData } = await supabase
    .from('country_data')
    .select('indicator_code, year, value, source')
    .eq('country_iso3', iso3)
    .order('year', { ascending: true });

  if (!tsData) { console.error('No data found'); return; }

  const rows: string[] = ['ISO3,country_name,year,indicator_code,value,unit,source'];
  for (const row of tsData) {
    rows.push([
      iso3,
      `"${country?.name || iso3}"`,
      row.year,
      row.indicator_code,
      row.value,
      """,  // unit from indicators table — enhance later
      `"${row.source || ''}"`,
    ].join(','));
  }

  const outPath = path.join(__dirname, '..', 'data', 'tableau', 'timeseries.csv');
  fs.writeFileSync(outPath, rows.join('\n'), 'utf-8');
  console.log(`Written ${rows.length - 1} rows to ${outPath}`);
}

// Main
const iso3Arg = process.argv[2];
if (iso3Arg) {
  pullTimeseries(iso3Arg.toUpperCase());
} else {
  pullCountriesLatest();
}
```

파일 #29: scripts/push-to-supabase.ts

```
/**
 * push-to-supabase.ts
 *
 * Pushes new indicator definitions and data to the existing Supabase database.
 * Used when adding EDGAR, IRENA, CAT, Net Zero Tracker data.
 *
 * Usage: npx tsx scripts/push-to-supabase.ts {action}
 *   - register-indicators: Add new indicator definitions
 *   - push-data {csv_path}: Push CSV data to country_data
```

```
 *
 * CSV format expected: ISO3,indicator_code,year,value,source
 */

import { createClient } from '@supabase/supabase-js';
import * as fs from 'fs';

const SUPABASE_URL = process.env.SUPABASE_URL!;
const SUPABASE_KEY = process.env.SUPABASE_SERVICE_KEY!;

const supabase = createClient(SUPABASE_URL, SUPABASE_KEY);

// New indicators to register (follows existing pattern)
const NEW_INDICATORS = [
  { source: 'EDGAR', code: 'EDGAR.GHG.TOTAL', name: 'Total GHG excl. LULUCF
(EDGAR)', unit: 'MtCO2eq', category: 'emissions', domain: 'emissions' },
  { source: 'EDGAR', code: 'EDGAR.CO2.FOSSIL', name: 'CO2 fossil fuels and industry
(EDGAR)', unit: 'MtCO2', category: 'emissions', domain: 'emissions' },
  { source: 'EDGAR', code: 'EDGAR.CH4.TOTAL', name: 'Total CH4 (EDGAR)', unit:
'MtCO2eq', category: 'emissions', domain: 'emissions' },
  { source: 'EDGAR', code: 'EDGAR.SECTOR.ENERGY', name: 'Energy sector share
(EDGAR)', unit: '%', category: 'emissions', domain: 'emissions' },
  { source: 'EDGAR', code: 'EDGAR.SECTOR.INDUSTRY', name: 'Industry sector share
(EDGAR)', unit: '%', category: 'emissions', domain: 'emissions' },
  { source: 'EDGAR', code: 'EDGAR.SECTOR.AGRICULTURE', name: 'Agriculture sector
share (EDGAR)', unit: '%', category: 'emissions', domain: 'emissions' },
  { source: 'EDGAR', code: 'EDGAR.SECTOR.WASTE', name: 'Waste sector share
(EDGAR)', unit: '%', category: 'emissions', domain: 'emissions' },
  { source: 'IRENA', code: 'IRENA.RE.CAPACITY', name: 'Renewable energy capacity', unit:
'GW', category: 'energy', domain: 'energy' },
  { source: 'IRENA', code: 'IRENA.RE.SHARE.ELEC', name: 'RE share of electricity
generation', unit: '%', category: 'energy', domain: 'energy' },
  { source: 'CAT', code: 'CAT.RATING', name: 'Climate Action Tracker rating', unit: 'text',
category: 'policy', domain: 'policy' },
  { source: 'NETZERO', code: 'NETZERO.YEAR', name: 'Net-zero target year', unit: 'year',
category: 'policy', domain: 'policy' },
  { source: 'NETZERO', code: 'NETZERO.LEGAL', name: 'Net-zero legal status', unit: 'text',
category: 'policy', domain: 'policy' },
  { source: 'NDC', code: 'NDC.3.SUBMITTED', name: 'NDC 3.0 submitted', unit: 'boolean',
category: 'policy', domain: 'policy' },
  { source: 'NDC', code: 'NDC.2035.TARGET', name: 'NDC 2035 target reduction %', unit:
'%', category: 'policy', domain: 'policy' },
];

async function registerIndicators() {
  console.log('Registering new indicators...');
  for (const ind of NEW_INDICATORS) {
    const { error } = await supabase
```

```
      .from('indicators')
      .upsert(ind, { onConflict: 'code' });
    if (error) {
      console.error(`Failed to register ${ind.code}:`, error.message);
    } else {
      console.log(`Registered: ${ind.code}`);
    }
  }
  console.log('Done. Total new indicators:', NEW_INDICATORS.length);
}

async function pushData(csvPath: string) {
  console.log(`Pushing data from ${csvPath}...`);
  const content = fs.readFileSync(csvPath, 'utf-8');
  const lines = content.trim().split('\n');
  const header = lines[0];
  console.log(`Header: ${header}`);
  console.log(`Rows: ${lines.length - 1}`);

  let success = 0;
  let failed = 0;

  // Process in batches of 500
  const batchSize = 500;
  const dataRows = lines.slice(1).map(line => {
    const [iso3, indicator_code, year, value, source] = line.split(',');
    return {
      country_iso3: iso3.trim(),
      indicator_code: indicator_code.trim(),
      year: parseInt(year.trim()),
      value: parseFloat(value.trim()),
      source: source?.trim().replace(/"/g, '') || null,
    };
  }).filter(r => !isNaN(r.value) && !isNaN(r.year));

  for (let i = 0; i < dataRows.length; i += batchSize) {
    const batch = dataRows.slice(i, i + batchSize);
    const { error } = await supabase
      .from('country_data')
      .upsert(batch, { onConflict: 'country_iso3,indicator_code,year' });
    if (error) {
      console.error(`Batch ${i}-${i + batch.length} failed:`, error.message);
      failed += batch.length;
    } else {
      success += batch.length;
    }
  }
```

```
  console.log(`Done. Success: ${success}, Failed: ${failed}`);
}

// Main
const action = process.argv[2];
if (action === 'register-indicators') {
  registerIndicators();
} else if (action === 'push-data') {
  const csvPath = process.argv[3];
  if (!csvPath) { console.error('Usage: push-to-supabase.ts push-data {csv_path}');
process.exit(1); }
  pushData(csvPath);
} else {
  console.log('Usage:');
  console.log('  npx tsx scripts/push-to-supabase.ts register-indicators');
  console.log('  npx tsx scripts/push-to-supabase.ts push-data {csv_path}');
}
```

파일 #30: docs/architecture.md

VisualClimate Trojan — System Architecture

두 레포 관계

```
┌──────────────────────────────────────────┐
┌──────────────────────────────────────────┐
│ Climate-RAG-for-ALL │ │ visualclimate-trojan │
│ (웹앱 프론트엔드) │ │ (분석·보고서·콘텐츠 엔진) │
│ │ │ │
│ Next.js + D3 + Vercel │ │ Claude Code + Tableau │
│ visualclimate.org │ │ reports + dashboards │
│ │ │ │
│ Supabase ←──── READ ────→ │←──→ │ READ + WRITE │
│ 172,121 rows │ │ +EDGAR, IRENA, NDC3.0 │
│ 250 countries │ │ +CAT, Net Zero Tracker │
│ 61 indicators │ │ │
└──────────────────────────────────────────┘
└──────────────────────────────────────────┘
│ │
↓↓
웹 대시보드 보고서 3종 (MD/PDF)
D3 차트 Tableau 대시보드 3종
포스터 PNG LinkedIn 콘텐츠
교육용 슬라이드
이메일 뉴스레터
```

데이터 흐름

Supabase (172,121행)
↓ pull-from-supabase.ts
data/tableau/countries_latest.csv + timeseries.csv
↓
Claude Skills (country-brief, comparison, ndc-gap)
↓
output/*.md (보고서)*
↓ *qa-checker*
output/.md (검증됨)
↓ content-repurposer
output/*-carousel.md, *-linkedin.md, etc.

동시에:
data/tableau/*.csv → Tableau Desktop → 대시보드 3종
tableau-spec-writer → 대시보드 설계도 → Harim이 Tableau에서 구현

Layer 구조
Layer	구성 요소	역할
Layer 1: 기반 자산	CLAUDE.md, schema.json, source-registry.json, citations.json, brand-config.md	모든 작업의 규칙과 데이터 정의
Layer 2: Claude 스킬	10개 스킬 (A1-A4, B1-B3, C1, D1, E1)	데이터를 분석하고 보고서를 생성
Layer 3: 출력물	보고서 3종, Tableau 3종, LinkedIn 콘텐츠	사용자에게 전달되는 최종 결과물

파일 #31: docs/claude-vs-tableau.md

Claude vs Tableau — 역할 분리 명세

원칙
Claude는 텍스트·분석·판단을 한다. Tableau는 시각화를 한다.
둘 사이의 데이터 교환은 CSV(data/tableau/)로만 이루어진다.

Claude가 하는 일
영역	구체적 작업	스킬
데이터 수집	Supabase 쿼리, API 호출, CSV 다운로드	pull-from-supabase.ts
데이터 정제	단위 변환, GWP 변환, 결측 처리	data-quality-checker
데이터 해석	스키마 매핑, 소스 비교, 용어 정의	schema-explainer, source-comparator, indicator-definer

분석	추세 계산, 갭 분석, 동료 비교	country-brief, ndc-gap-brief
보고서 생성	마크다운 보고서 작성	country-brief, comparison-brief, ndc-gap-brief
검증	사실 확인, 단위 일관성, 출처 검증	qa-checker
콘텐츠 변환	캐러셀, 텍스트 포스트, 슬라이드	content-repurposer
Tableau 설계	대시보드 스펙, 계산식 설계	tableau-spec-writer
CSV 생성	countries_latest.csv, timeseries.csv	scripts, country-brief

Tableau가 하는 일
영역	구체적 작업	입력
CSV 읽기	data/tableau/*.csv 연결	Claude가 생성한 CSV
차트 렌더링	선, 막대, 파이, 지도, 스캐터	CSV 데이터
계산 필드 실행	calculated-fields.md의 공식	Claude가 설계한 공식
대시보드 레이아웃	시트 배치, 필터, 파라미터	Claude의 tableau-spec
인터랙티브	필터 액션, 하이라이트, 툴팁	tableau-spec
내보내기	PDF, 이미지, Tableau Public	완성된 대시보드

Claude가 절대 하지 않는 것
- 차트 이미지 직접 생성 (Tableau가 함)
- 대시보드 레이아웃 직접 구현 (설계만 함)
- Tableau 파일(.twbx) 직접 수정

Tableau가 절대 하지 않는 것
- 데이터 수집/ETL (Claude/스크립트가 함)
- 텍스트 분석/보고서 작성 (Claude가 함)
- 데이터 품질 검증 (Claude가 함)
- 콘텐츠 변환 (Claude가 함)

연결 지점 (CSV 규격)
Claude가 이 CSV를 만든다 → Tableau가 이 CSV를 읽는다
- data/tableau/countries_latest.csv: 250행 × 15열 (최신 값)
- data/tableau/timeseries.csv: N행 × 7열 (연도별)
- 열 이름은 schema.json의 tableau_field 값 사용

파일 #32: docs/content-strategy.md

콘텐츠 전략: 정보성 vs 제품

원칙
같은 스킬, 같은 데이터. 정보성에는 "과정 + 일부"를, 제품에는 "전체 완성본"을 넣는다.

정보성 콘텐츠 (무료, LinkedIn 배포)
콘텐츠	만드는 스킬	LinkedIn 형태	목적
소스 비교표	source-comparator	캐러셀 4장	Save + 팔로우
용어 정의 + 체크리스트	indicator-definer	캐러셀 5장	Save + 전문성

데이터 품질 Before/After	data-quality-checker	이미지 1장 + 텍스트	교육
Country Brief Key Metrics만	country-brief (일부)	이미지 1장	제품 맛보기
Tableau 차트 해설	tableau-spec-writer + Tableau	이미지 1장 + 텍스트	시각화 역량
NDC Gap 한 줄 요약	ndc-gap-brief (일부)	텍스트 포스트	시의성

제품 (유료 또는 리드 생성)
제품	만드는 스킬	전달 형태	수익화
Country Climate Brief 전문	country-brief	PDF (이메일 수집)	리드
Comparison Brief	comparison-brief	PDF (이메일 수집)	리드
NDC Gap Brief	ndc-gap-brief	PDF ($49/국가)	직접 수익
24주 브리프 묶음	위 3개 반복	PDF 번들	직접 수익
Tableau 대시보드	tableau-spec-writer + Tableau	Tableau Public + 임베드	트래픽
이 시스템 구축 강의	전체 과정 문서화	영상/PDF	직접 수익
맞춤 컨설팅	전체 스킬 활용	보고서 + 대시보드	$10K+

주간 콘텐츠 캘린더 (예시)
요일	유형	스킬	예시
월	소스 비교	source-comparator	"EDGAR vs Climate TRACE: 왜 숫자가 다른가?"
화	데이터 드롭	country-brief 일부	"한국 1인당 배출 11.4t — 세계 평균의 2.3배"
수	국가 스포트라이트	country-brief 캐러셀	"Germany Country Brief 핵심 요약"
목	방법론	indicator-definer	"GHG Intensity vs Carbon Intensity — 헷갈리는 이유"
금	비교	comparison-brief 캐러셀	"한국 vs 일본 vs 독일: 누가 앞서나?"

파일 #33: brand-config.md

VisualClimate — Brand Configuration

Colors
Name	Hex	Use
Primary	#0066FF	Links, CTAs, main accent
Changer (Good)	#00A67E	Positive metrics, On Track
Starter (Warning)	#F59E0B	Mixed metrics, Narrow Gap
Talker (Bad)	#E5484D	Negative metrics, Significant Gap
Critical	#7F1D1D	Worst case, Critical Gap
Background	#FFFFFF	All backgrounds
Section BG	#F8F9FA	Alternating sections
Text Primary	#1A1A2E	Headings, body text
Text Secondary	#4A4A6A	Subtitles, labels
Text Muted	#8888A0	Captions, footnotes
Border	#E5E7EB	Dividers, card borders
Data Missing	#CCCCCC	[DATA MISSING] indicators

Typography

Use	Font	Weight	Size
Headings	Inter	Bold (700)	14-24pt
Body	Inter	Regular (400)	11-14pt
Data/Code	JetBrains Mono	Regular	10-12pt
Captions	Inter	Regular	9-10pt

Logo
- Position: bottom-right on all outputs
- Size: 80 × 20 px on carousel slides
- Text-only fallback: "VisualClimate" in Inter Bold #0066FF

Report Header
Every report starts with:

{Report Title}

Data vintage: {year} | **Generated**: {date} | **Engine**: VisualClimate Trojan v1.0

Carousel Slide Rules
- 1080 × 1080 px
- White background (#FFFFFF)
- Max 2 colors per slide (Primary + one accent)
- Source citation: bottom-left, 9pt, #8888A0
- Logo: bottom-right

Prohibited
- Dark backgrounds (bg-slate-900, #0a0a1a, etc.)
- More than 3 accent colors per piece
- Stock photos
- Emojis in reports (ok in LinkedIn captions, max 2)

파일 #34: .env.example

```
# Supabase — existing VisualClimate database
SUPABASE_URL=https://your-project.supabase.co
SUPABASE_SERVICE_KEY=eyJ...your-service-key

# Optional: for future API integrations
CLIMATE_TRACE_API_KEY=
OPENAI_API_KEY=
```

파일 #35: package.json

```
{
  "name": "visualclimate-trojan",
  "version": "1.0.0",
  "description": "Climate data analysis, report generation, and content engine for
VisualClimate",
  "scripts": {
    "pull": "npx tsx scripts/pull-from-supabase.ts",
    "pull:country": "npx tsx scripts/pull-from-supabase.ts",
    "push:indicators": "npx tsx scripts/push-to-supabase.ts register-indicators",
    "push:data": "npx tsx scripts/push-to-supabase.ts push-data"
  },
  "dependencies": {
    "@supabase/supabase-js": "^2.45.0"
  },
  "devDependencies": {
    "tsx": "^4.7.0",
    "typescript": "^5.4.0",
    "@types/node": "^20.11.0"
  }
}
```

파일 #36: [README.md]

```
# VisualClimate Trojan

Climate data analysis, report generation, and content engine for
[VisualClimate](https://visualclimate.org).

## What This Does
- Reads from existing Supabase (172,121 rows, 250 countries, 61 indicators)
- Adds new data sources (EDGAR, IRENA, CAT, Net Zero Tracker)
- Generates 3 types of climate briefs (Country, Comparison, NDC Gap)
- Designs Tableau dashboard specifications
- Transforms reports into LinkedIn content
- Quality-checks everything before publishing

## Quick Start
```bash
# 1. Clone and install
git clone https://github.com/HarimJung/visualclimate-trojan.git
cd visualclimate-trojan
```
```

```
npm install
cp .env.example .env
# Fill in SUPABASE_URL and SUPABASE_SERVICE_KEY

# 2. Pull existing data
npm run pull

# 3. Start Claude Code
claude
# Claude reads CLAUDE.md automatically

# 4. Generate first brief
/country-brief KOR

# 5. Quality check
/qa output/KOR-country-brief-2026-04-07.md

# 6. Make LinkedIn content
/repurpose output/KOR-country-brief-2026-04-07.md linkedin-carousel

# 7. Get Tableau specs
/tableau-spec country-overview
```

Architecture

See docs/architecture.md

Skills (10)

Group	Skill	Trigger
A: Data Interpretation	schema-explainer	/explain-schema
A: Data Interpretation	source-comparator	/compare-sources {topic}
A: Data Interpretation	data-quality-checker	/check-quality
A: Data Interpretation	indicator-definer	/define {terms}
B: Report Generation	country-brief	/country-brief {ISO3}
B: Report Generation	comparison-brief	/compare-countries {ISO3s}
B: Report Generation	ndc-gap-brief	/ndc-gap {ISO3}
C: Verification	qa-checker	/qa {path}

D: Content	content-repurposer	/repurpose {path} {format}
E: Tableau	tableau-spec-writer	/tableau-spec {type}

이게 전부입니다. 36개 파일, 빠진 것 없습니다.

Claude Code에서 실행하는 순서:

1. `mkdir visualclimate-trojan && cd visualclimate-trojan`
2. 위 파일들을 각 경로에 생성
3. `npm install`
4. `.env` 파일에 기존 Supabase 키 입력
5. `npm run pull` — 기존 데이터 CSV로 추출
6. `claude` — Claude Code 실행 (CLAUDE.md 자동 로드)
7. `/country-brief KOR` — 첫 번째 보고서 생성

Claude가 보고서를 만들면 Tableau CSV도 동시에 업데이트되고, 하림은 Tableau에서 `tableau-spec-writer`가 설계한 대로 대시보드를 만들면 됩니다. 각 파트가 독립적으로 관리 가능하고, 국가명만 바꾸면 반복 생산됩니다.